香港中文大學統計學系
Department of Statistics
THE CHINESE UNIVERSITY OF HONG KONG

STAT6207
Applied Deep Learning

L3, 25 Sep 2026
By Dr. Ming Cheung



Announcement

- Midterm exam (papers + programming) will be on 23 Oct 26
- Location: _____
- There will be two parts
  - Written: MC and short question
  - Programming: You can use your computer and agent coder
- No communication is allowed, but you can bring a cheat sheet
- Make sure that you can run all demo and assignments on your computer
- Duration 2.5 hours, you will need to complete the written part first
- You will know the questions in Programming part at the beginning



Selected (last year)

Similarity: 0.75



Similarity: 0.74



Similarity: 0.74



Similarity: 0.72



Similarity: 0.72



Similarity: -0.25



Similarity: -0.17



Similarity: -0.10

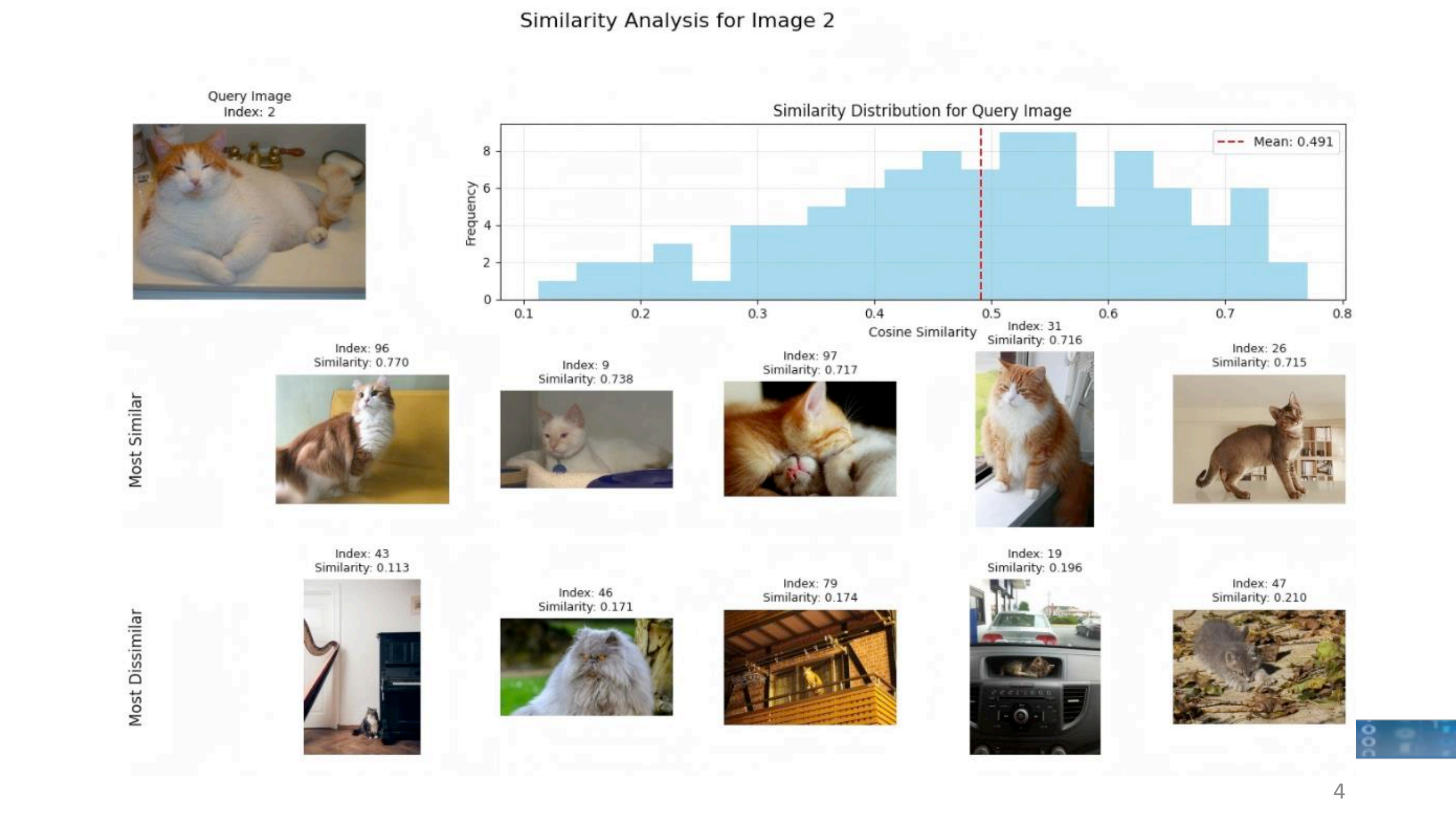


Similarity: -0.01



Similarity: 0.01



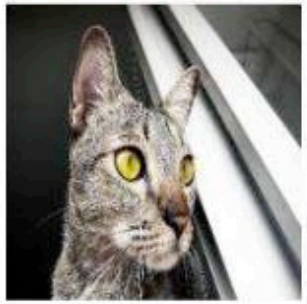

Similarity Analysis for Image 2
Query Image
Index: 2
Similarity Distribution for Query Image
Mean: 0.491
Frequency
Cosine Similarity
0
2
4
6
8
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Most Similar
Index: 96
Similarity: 0.770
Index: 9
Similarity: 0.738
Index: 97
Similarity: 0.717
Index: 31
Similarity: 0.716
Index: 26
Similarity: 0.715
Most Dissimilar
Index: 43
Similarity: 0.113
Index: 46
Similarity: 0.171
Index: 79
Similarity: 0.174
Index: 19
Similarity: 0.196
Index: 47
Similarity: 0.210

Visualize performance on 10 unseen images

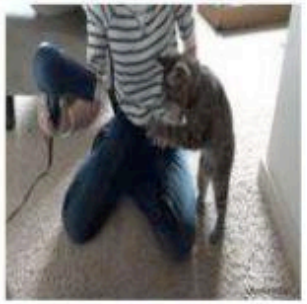
True: Cat
Pred: Cat



True: Cat
Pred: Cat



True: Cat
Pred: Dog



True: Cat
Pred: Cat



True: Cat
Pred: Cat



True: Cat
Pred: Dog



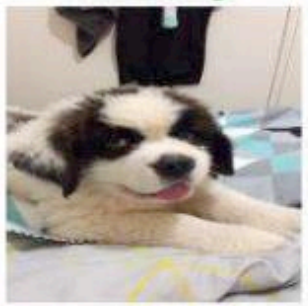
True: Cat
Pred: Cat



True: Dog
Pred: Cat



True: Dog
Pred: Dog



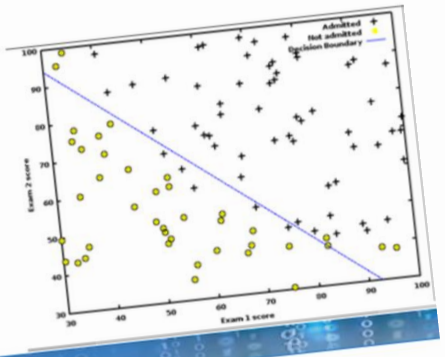
True: Cat
Pred: Cat



Last Lecture

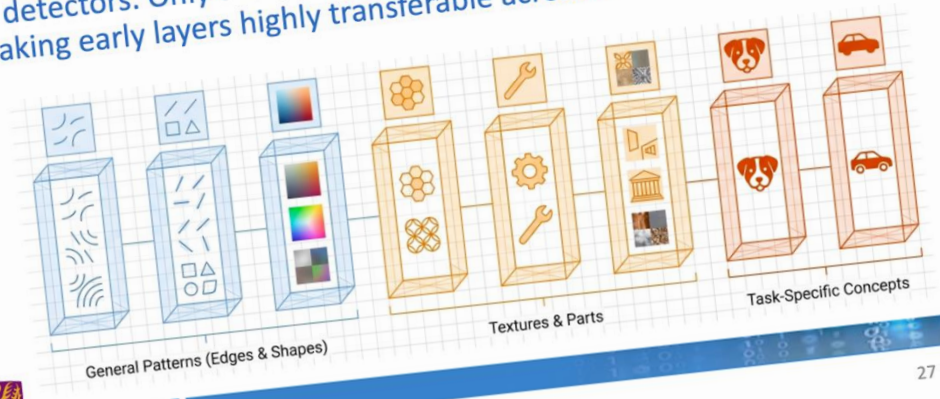
Classification: Decision Rule

- To Classify instances, we obtain a point estimation of y:
  - $\hat{y}$=argmax $p(y|x,w)$
- In other words, the rules is: $\hat{y}$=1 iff $p(y=1|x,w)>0$
- This is called the optimal Bayes classifier
  - e.g., $\hat{y}$=1 iff $w^T x>0$ as a linear classifier



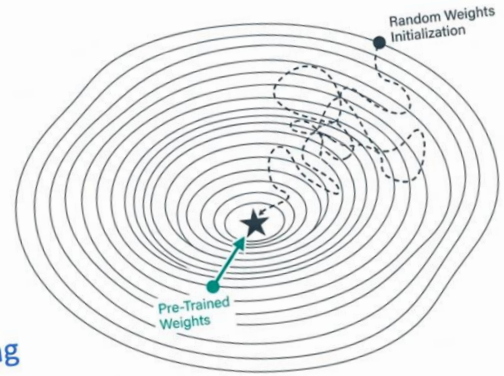
The General-to-Specific Spectrum of Deep Learning

- Deep networks learn hierarchical features. Early layers act as universal feature detectors. Only the deepest layers specialize in the original training task, making early layers highly transferable across domains



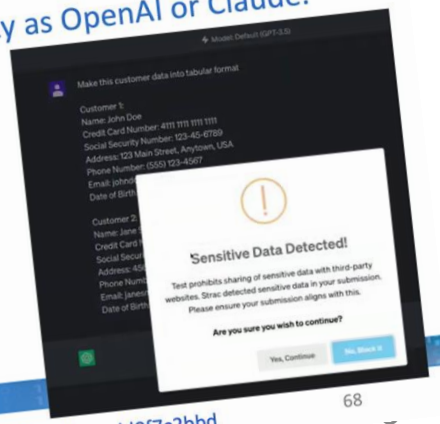
The Paradigm Shift: Standing on the Shoulders of Giants

- Instead of training from scratch, transfer learning takes a model already trained on a massive dataset (like ImageNet for vision, or BERT for text) and uses its learned features as a starting point.
- The Result: 80-90% reduction in required training data and compute costs. You bypass the phase of teaching the network basic reality.



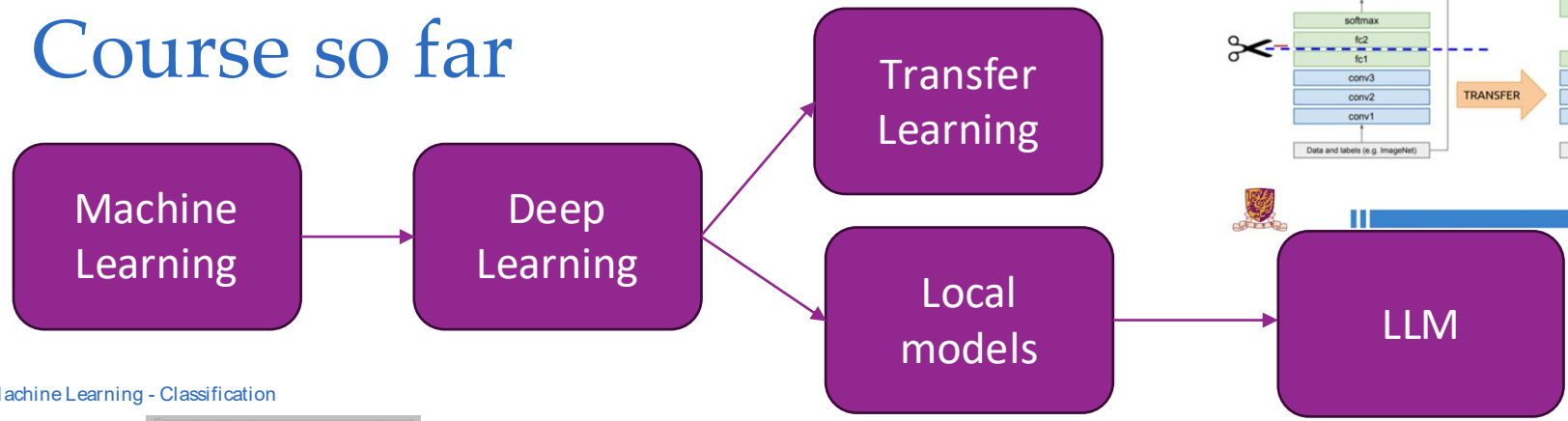
Why Local LLMs?

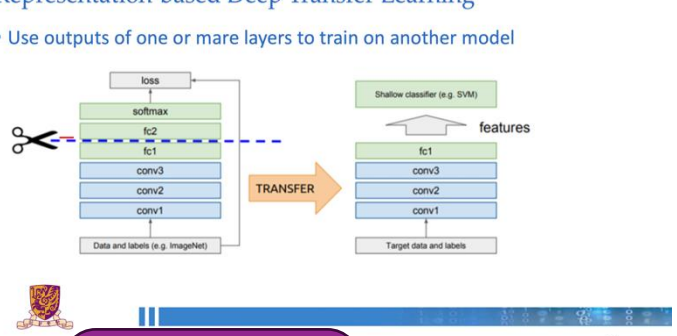
- **Privacy**: You can prompt local LLMs in a multi-turn manner without your prompt data leaving your localhost.
- **Support and Security**: similar support and security as OpenAI or Claude.
- **Offline Support:** Use LLM while offline.

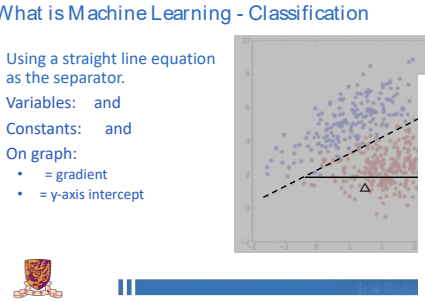


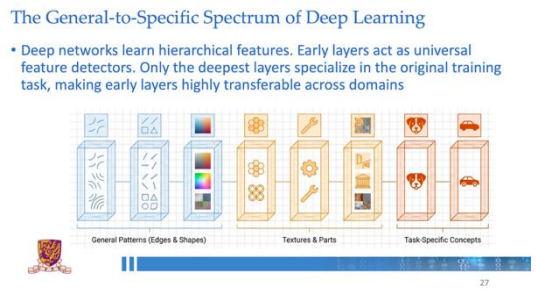
/@osgyamfi/the-6-best-llm-tools-to-run-models-locally-eedd0f7c2bbd

Course so far









Other options: Ollama

```python
import ollama

def generate_response():
    try:
        response = ollama.chat(
            model='llama3',  # Replace with 'mistral', 'phi3', or whatever model you have
            messages=[
                {
                    'role': 'user',
                    'content': 'Why is the sky blue? Give me a short, one-sentence answer',
                },
            ]
        )
        # Access the response content
        print("Response:")
        print(response['message']['content'])

    except Exception as e:
        print(f"An error occurred: {e}")

if __name__ == "__main__":
    generate_response()
```

73

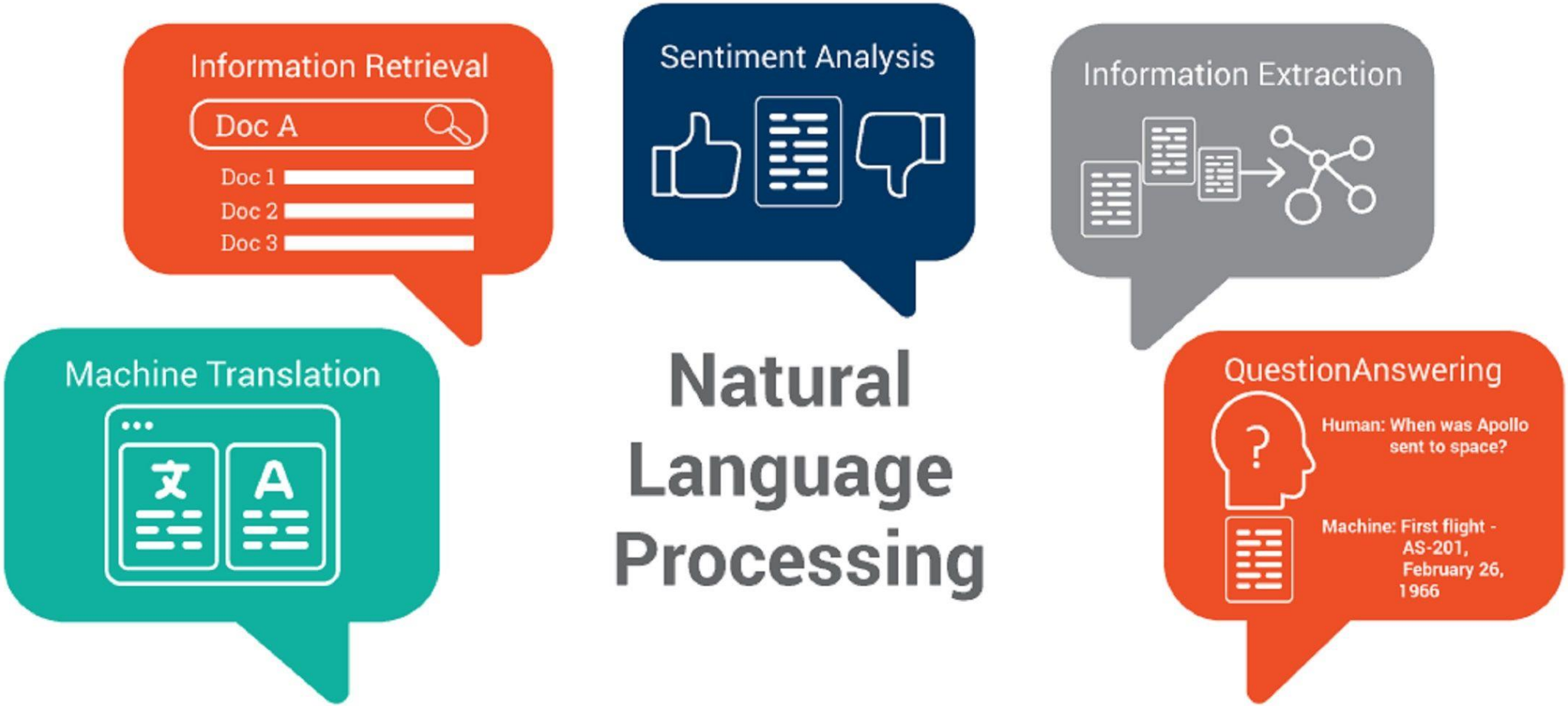
How do we process textual data?

Agenda

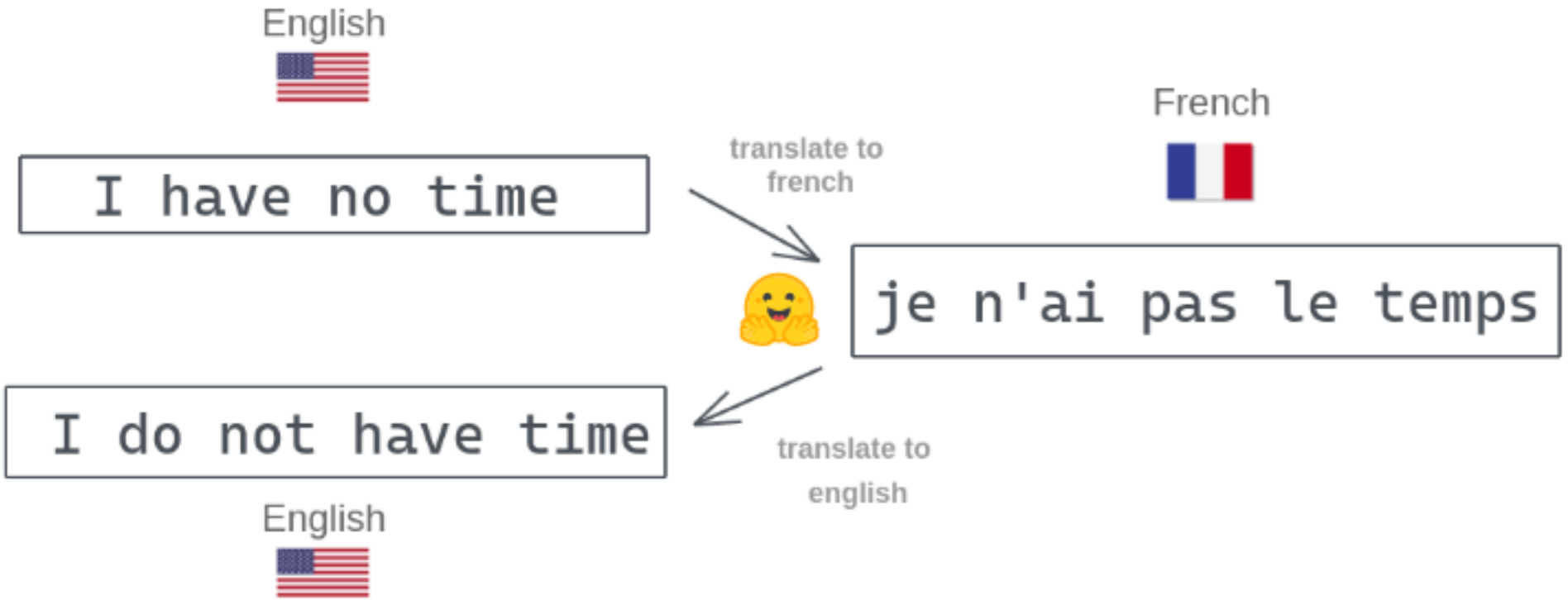
- What is NLP and Large Language Model (LLM)
- Retrieval Augmented Generation (RAG)
- Agentic AI



Natural Language Processing



How to understand Language model

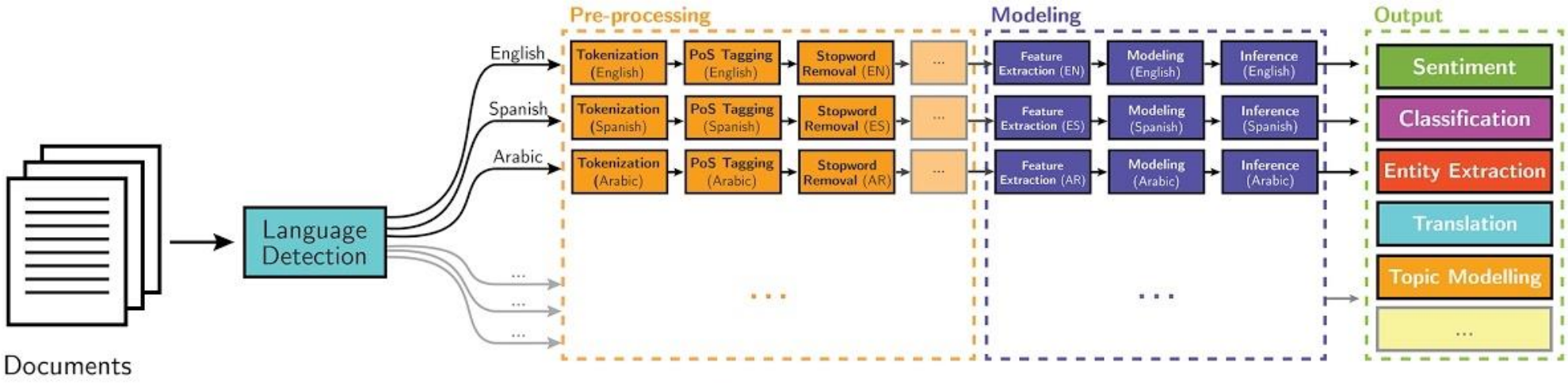


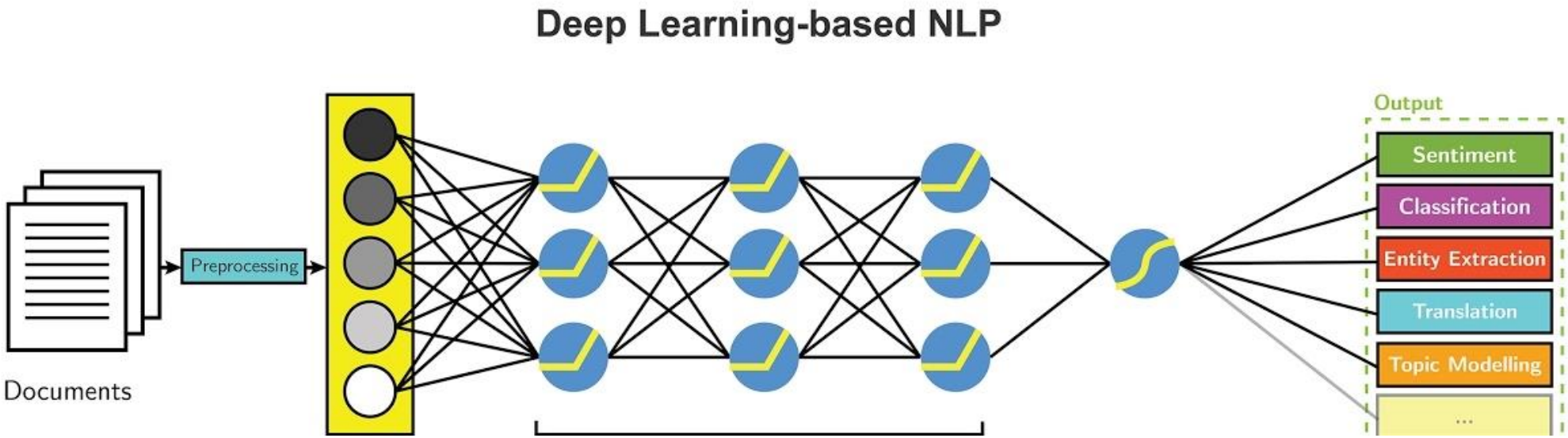
Retrieving data (Named Entity Recognition)

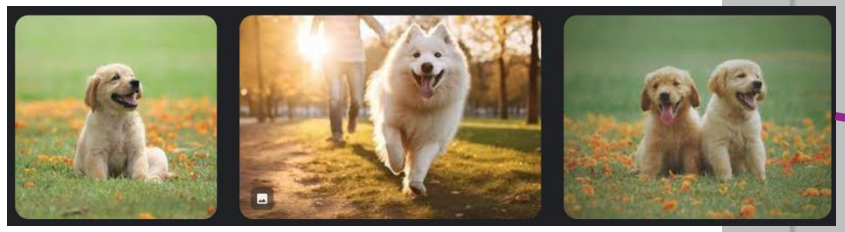
contentSkip to site indexPoliticsSubscribeLog InSubscribeLog InToday's PaperAdvertisementSupported ORG byF.B.I. Agent Peter Strzok PERSON , Who Criticized Trump PERSON in Texts, Is FiredImagePeter Strzok, a top F.B.I. GPE counterintelligence agent who was taken off the special counsel investigation after his disparaging texts about President Trump PERSON were uncovered, was fired. CreditT.J. Kirkpatrick PERSON for The New York TimesBy Adam Goldman ORG and Michael S. SchmidtAug PERSON . 13 CARDINAL , 2018WASHINGTON CARDINAL — Peter Strzok PERSON , the F.B.I. GPE senior counterintelligence agent who disparaged President Trump PERSON in inflammatory text messages and helped oversee the Hillary Clinton PERSON email and Russia GPE investigations, has been fired for violating bureau policies, Mr. Strzok PERSON 's lawyer said Monday DATE .Mr. Trump and his allies seized on the texts — exchanged during the 2016 DATE campaign with a former F.B.I. GPE lawyer, Lisa Page — in PERSON assailing the Russia GPE investigation as an illegitimate "witch hunt." Mr. Strzok PERSON , who rose over 20 years DATE at the F.B.I. GPE to become one of its most experienced counterintelligence agents, was a key figure in the early months DATE of the inquiry.Along with writing the texts, Mr. Strzok PERSON was accused of sending a highly sensitive search warrant to his personal email account.The F.B.I. GPE had been under immense political pressure by Mr. Trump PERSON to dismiss Mr. Strzok PERSON , who was removed last summer DATE from the staff of the special counsel, Robert S. Mueller III PERSON . The president has repeatedly denounced Mr. Strzok PERSON in posts on



Conventional vs. Deep Learning



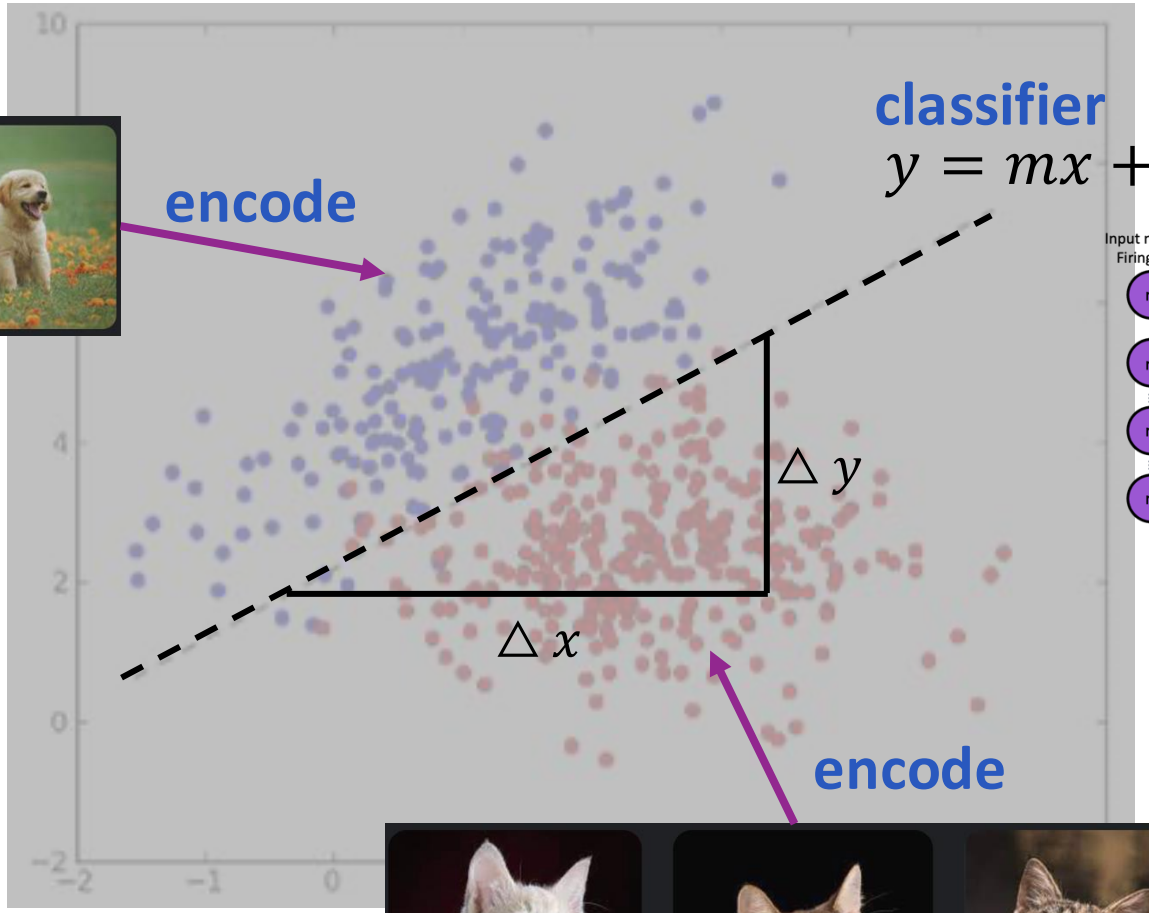


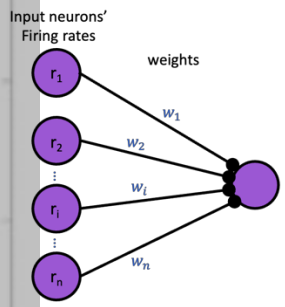


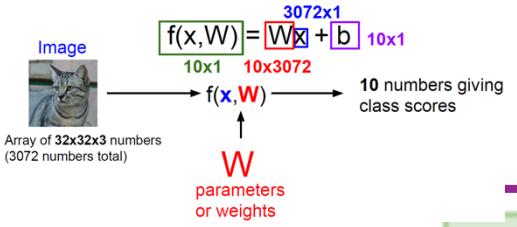
encode

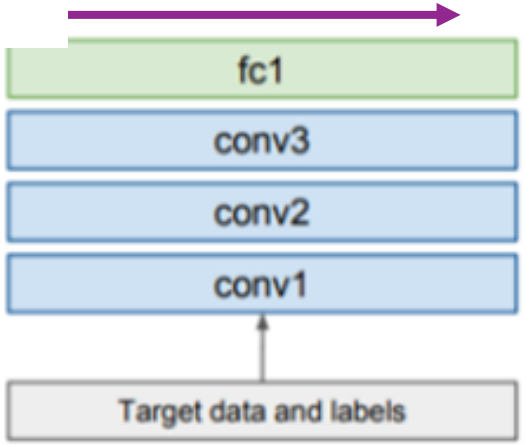
classifier

$y = mx + c$



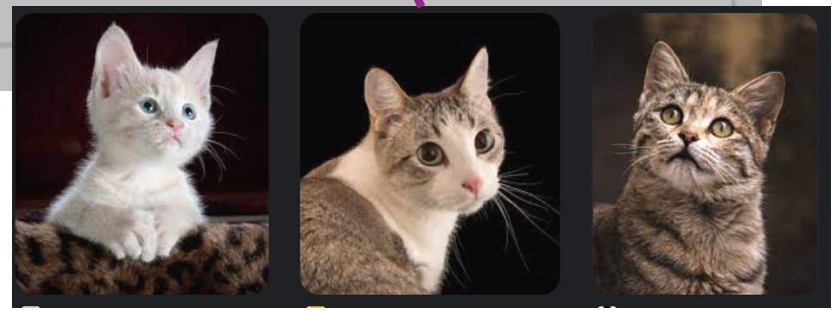




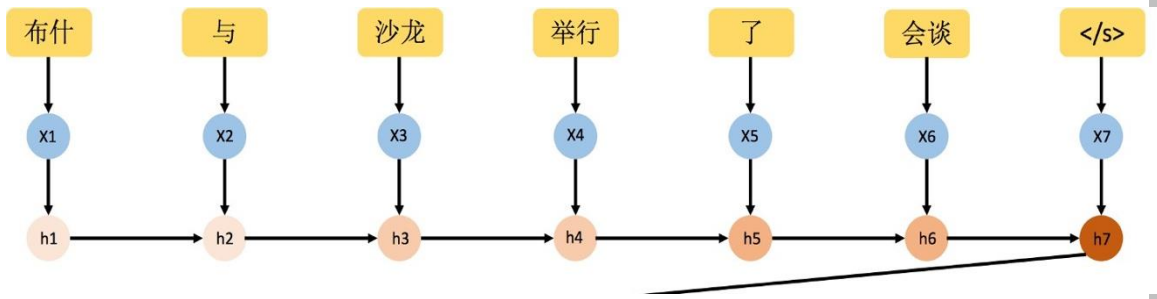


Pre-train model (encoder)

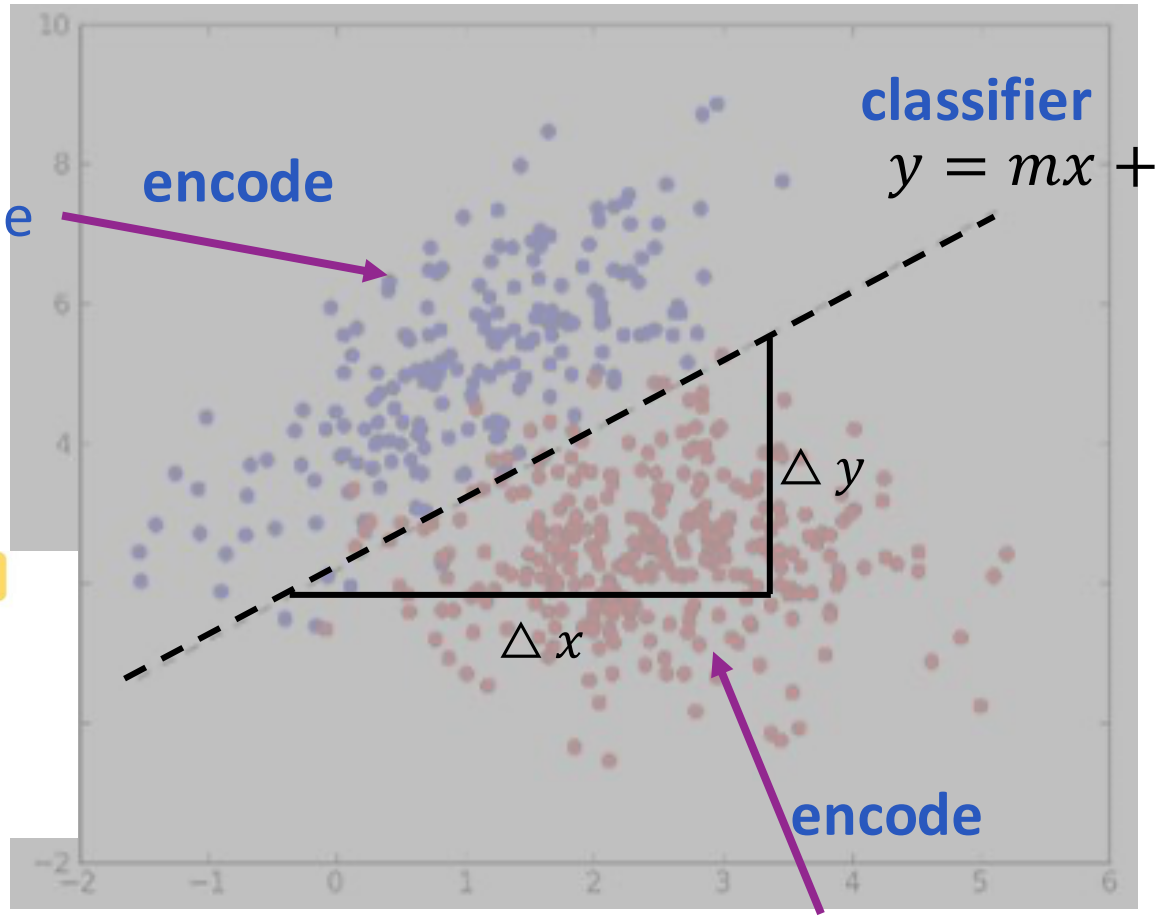
encode



This is a good movie
I highly recommend this movie
I will watch it again and again



Pre-train model (encoder)

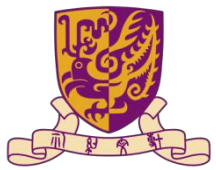


This is not a good movie
I won't recommend this movie
I will not watch it again

Recall: One Hot Vectors

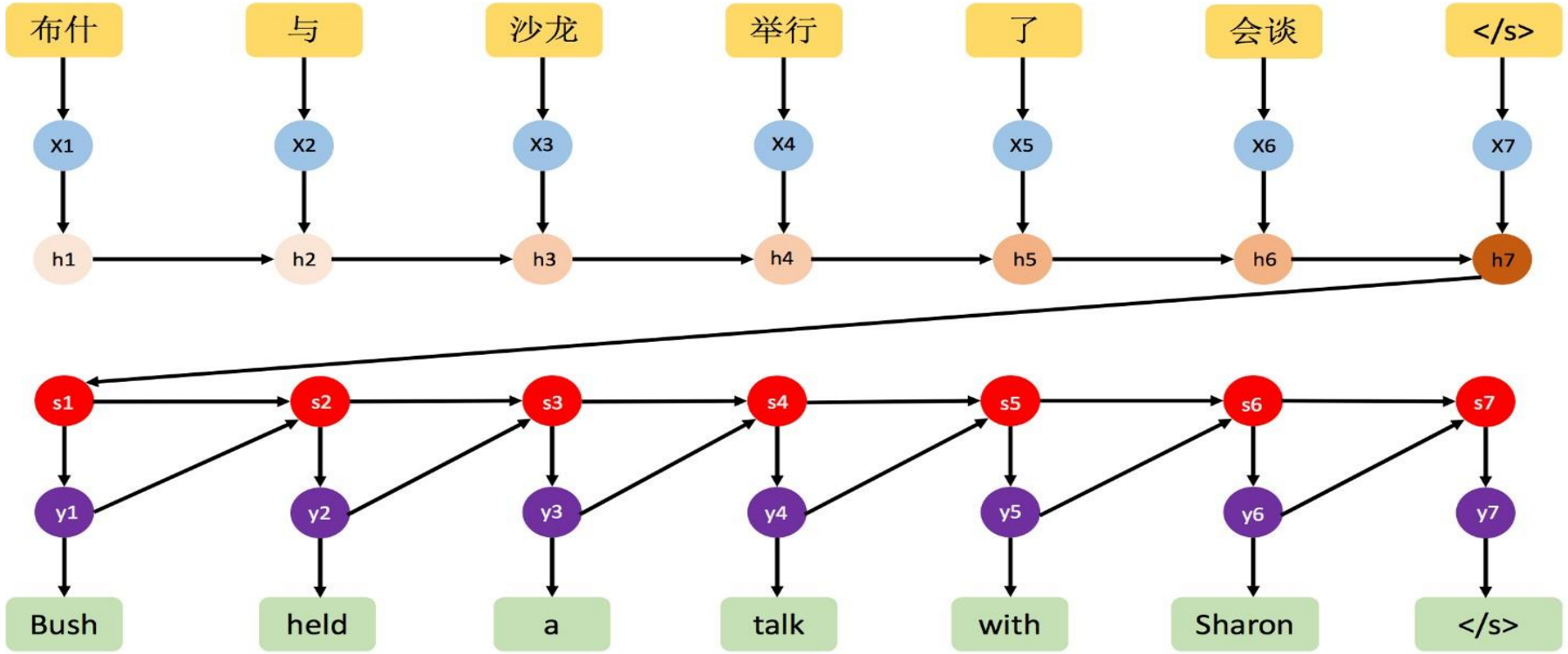
- A binary vector used to represent categorical /text data
- Each category is represented by a unique index in the vector.
- without any implicit relationships between categories.

Island		Biscoe	Dream	Torgensen
Biscoe	→	1	0	0
Torgensen		0	0	1
Dream		0	1	0



Seq2seq for machine translation

- The generated sequence can be a translation of the input sequence in machine translation



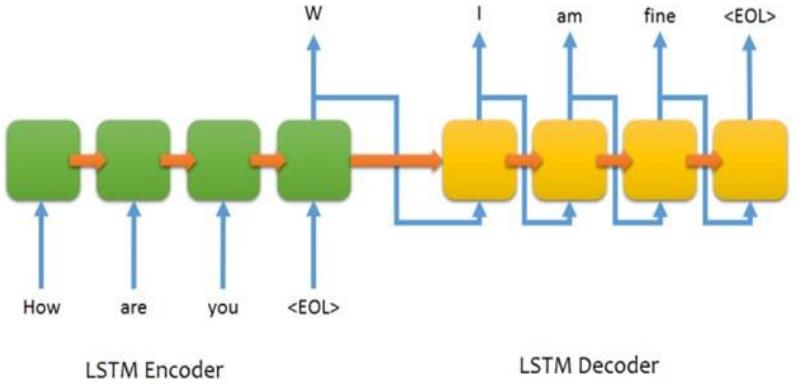
(Sutskever et al., 2014)

Recall: Seq2seq for text generation

- Models for labelled data:

$$\{x_i, y_i\}_{i=1}^{N} \rightarrow P(y|x)$$

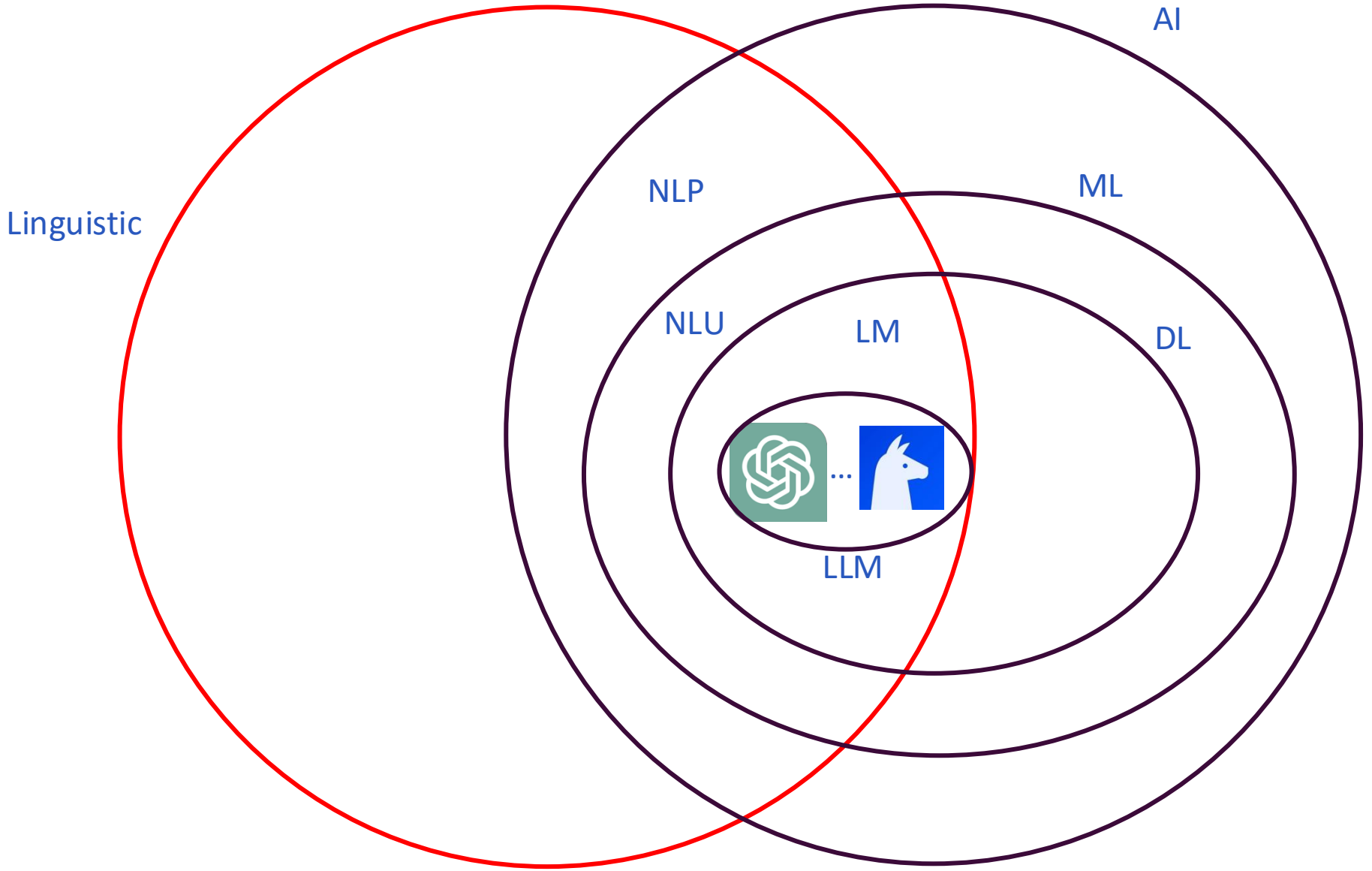
where each sample is an **input** x_i and **an output** y_i.



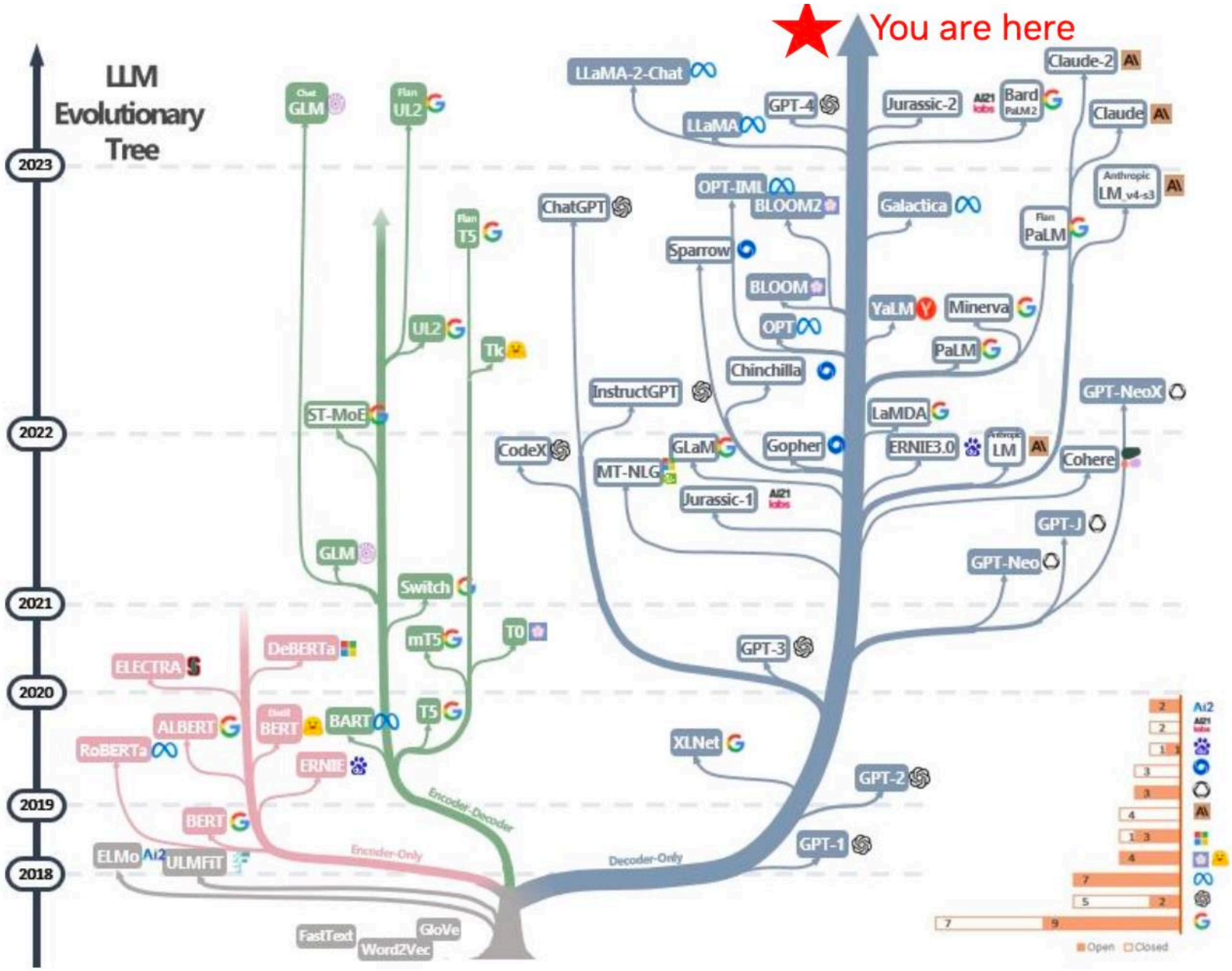
- Each training example consists of **a sequence of inputs**
  - $(\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(\tau i)})$ and **a sequence of outputs** $(\mathbf{y}^{(1)}, \ldots, \mathbf{y}^{(\tau i)})$,
  - (*"How", "are", "you"*, –, –, –), (–, –, –, *"I", "am", "fine"*).

- Training data can be from dialogue or machinetranslation




AI
Linguistic
NLP
ML
NLU
LM
DL
...
LLM

LLM



From: ***Harnessing the Power of LLMs in Practice: A Survey on ChatGPT and Beyond***
Jingfeng Yang, Hongye Jin, Ruixiang Tang, Xiaotian Han, Qizhang Feng, Haoming Jiang, Bing Yin, Xia Hu
https://github.com/Mooler0410/LLMsPracticalGuide

Attention Is All You Need

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Niki Parmar* Google Research nikip@google.com

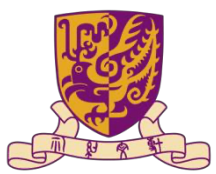
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Łukasz Kaiser* Google Brain lukaszkaiser@google.com

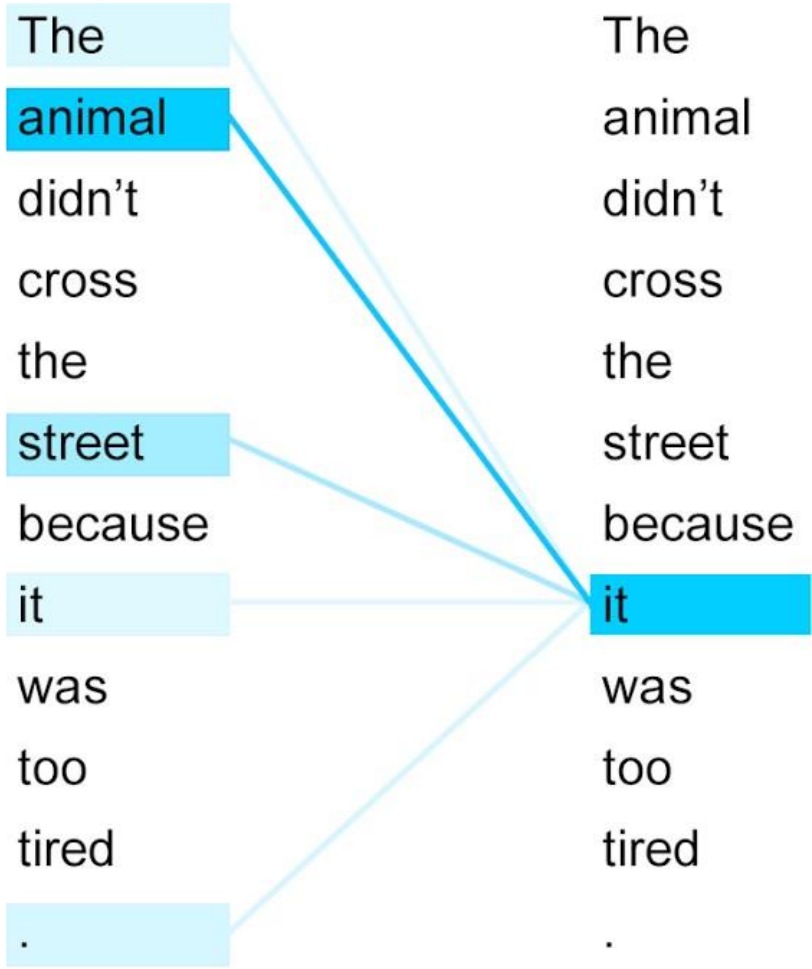
Illia Polosukhin* ‡ illia.polosukhin@gmail.com



Self-attention Layer (More on this later)

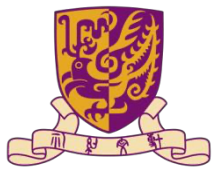
- "improve" the representation using other tokens

- Example:
  - What does "it" refer to? The street or the animal?
  - Self-attention: associate "it" with "animal"

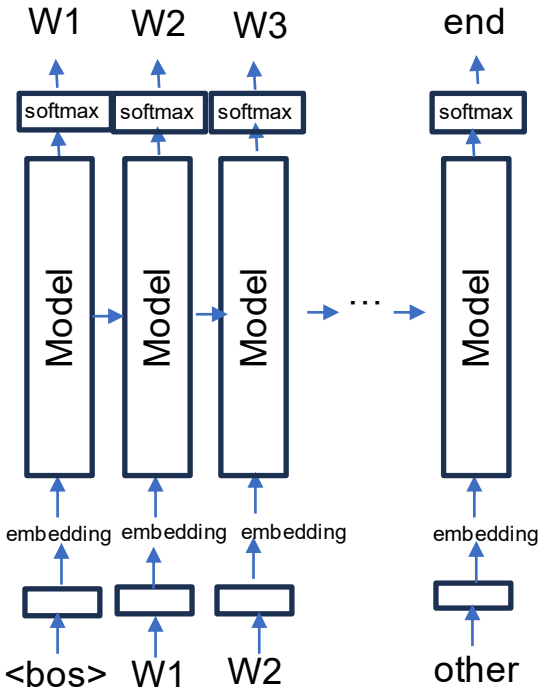


LLM: Model Prediction

- It works on arbitrary text, even text that is not in the tokeniser's training data
- It attempts to let the model see common subwords. For instance, "ing" is a common subword in English, so BPE encodings will often split "encoding" into tokens like "encod" and "ing" (instead of e.g. "enc" and "oding").
- Because the model will then see the "ing" token again and again in different contexts, it helps models generalise and better understand grammar.



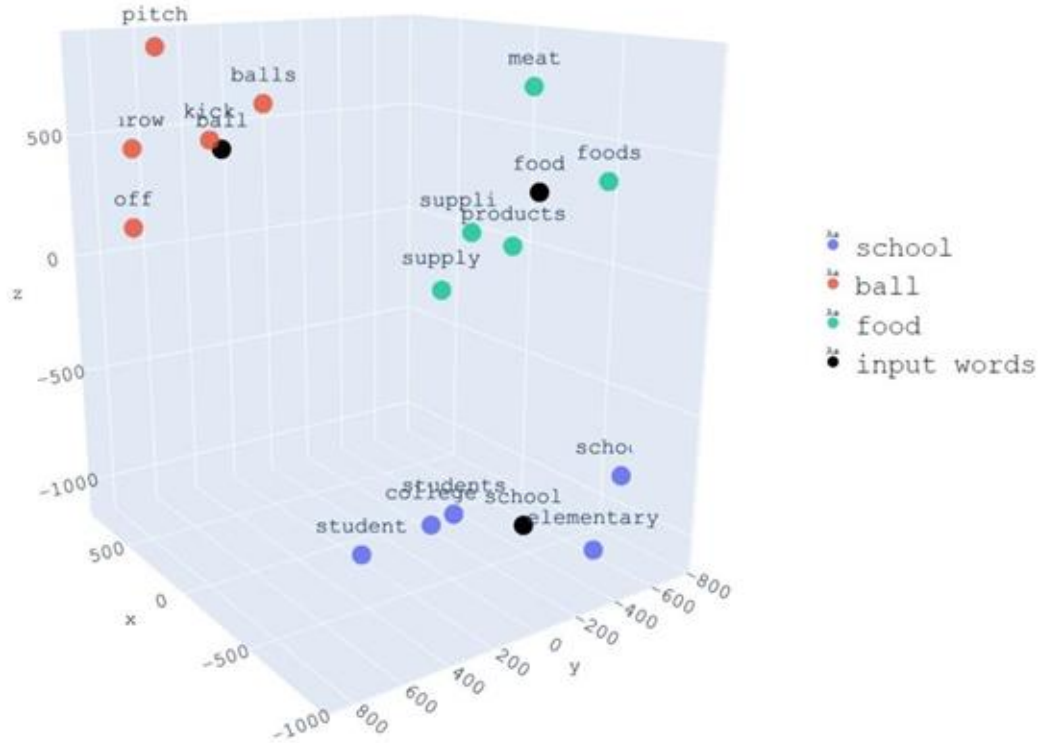
LLM: Model Prediction



Next word prediction is inherently an unsupervised task

Represent semantic distance through spatial distance

Word Embedding



LLM: Model Prediction

Process

"Many words don't map to one token: indivisible."

⬇ Tokenization

```
Unicode characters like emojis may be split.
[7085, 2456, 836, 470, 3975, 284, 530, 11241, 25, 773, 452, 12843, 13]
```

⬇ Embedding

2.3	-3.2	8.3	5.4	2.1	3.9	-8.9	3.8	3.9	3.3
4.5	5.9	4.5	7.1	1.0	5.3	5.0	3.1	0.7	5.0
...	...	...	...	...	...	...	...	...	...
3.8	1.2	3.8	9.0	9.3	3.1	4.2	0.8	9.2	5.8

[d_model, length]

⬇ N Transformer layers

3.2	-2.3	3.8	4.5	1.2	9.3	-9.8	8.3	9.3	3.3
5.4	9.5	5.4	1.7	0.1	3.5	0.5	1.3	7.0	0.5
...	...	...	...	...	...	...	...	...	...
8.3	2.1	8.3	0.9	3.9	1.3	2.4	8.0	2.9	8.5

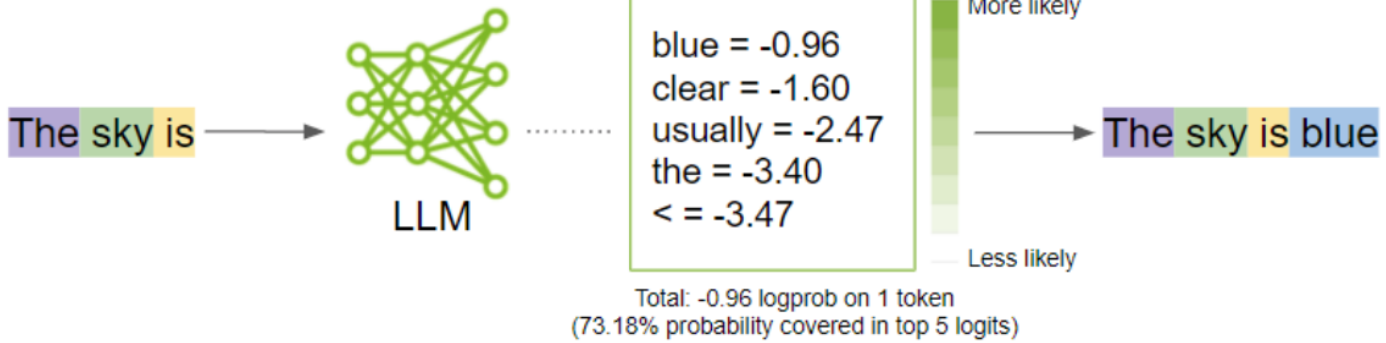
[d_model, length]

⬇ Loss function (predict next token given previous)



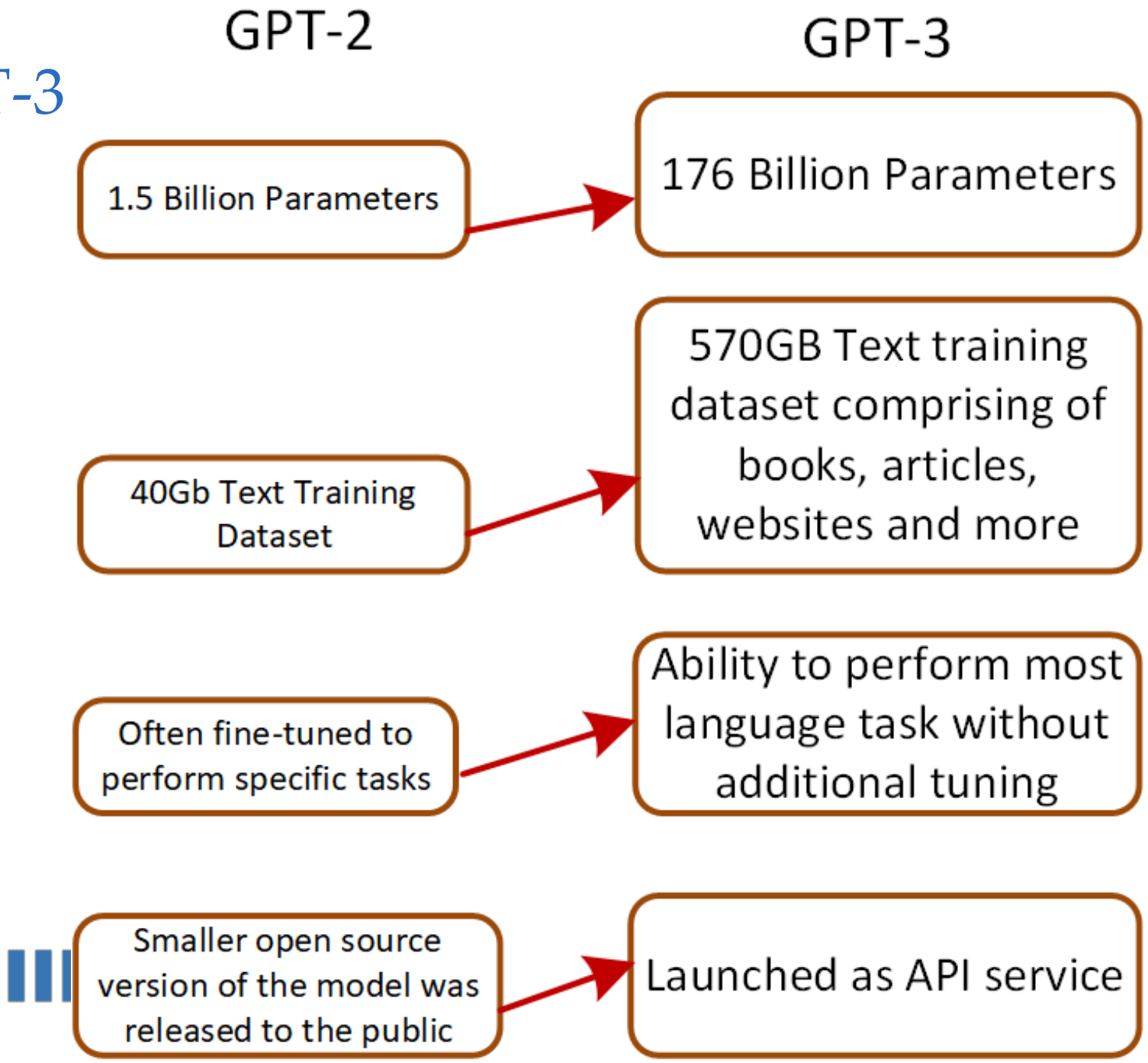
2.6

[]





GPT-2 to GPT-3


GPT-2
GPT-3
1.5 Billion Parameters
176 Billion Parameters
40Gb Text Training Dataset
570GB Text training dataset comprising of books, articles, websites and more
Often fine-tuned to perform specific tasks
Ability to perform most language task without additional tuning
Smaller open source version of the model was released to the public
Launched as API service

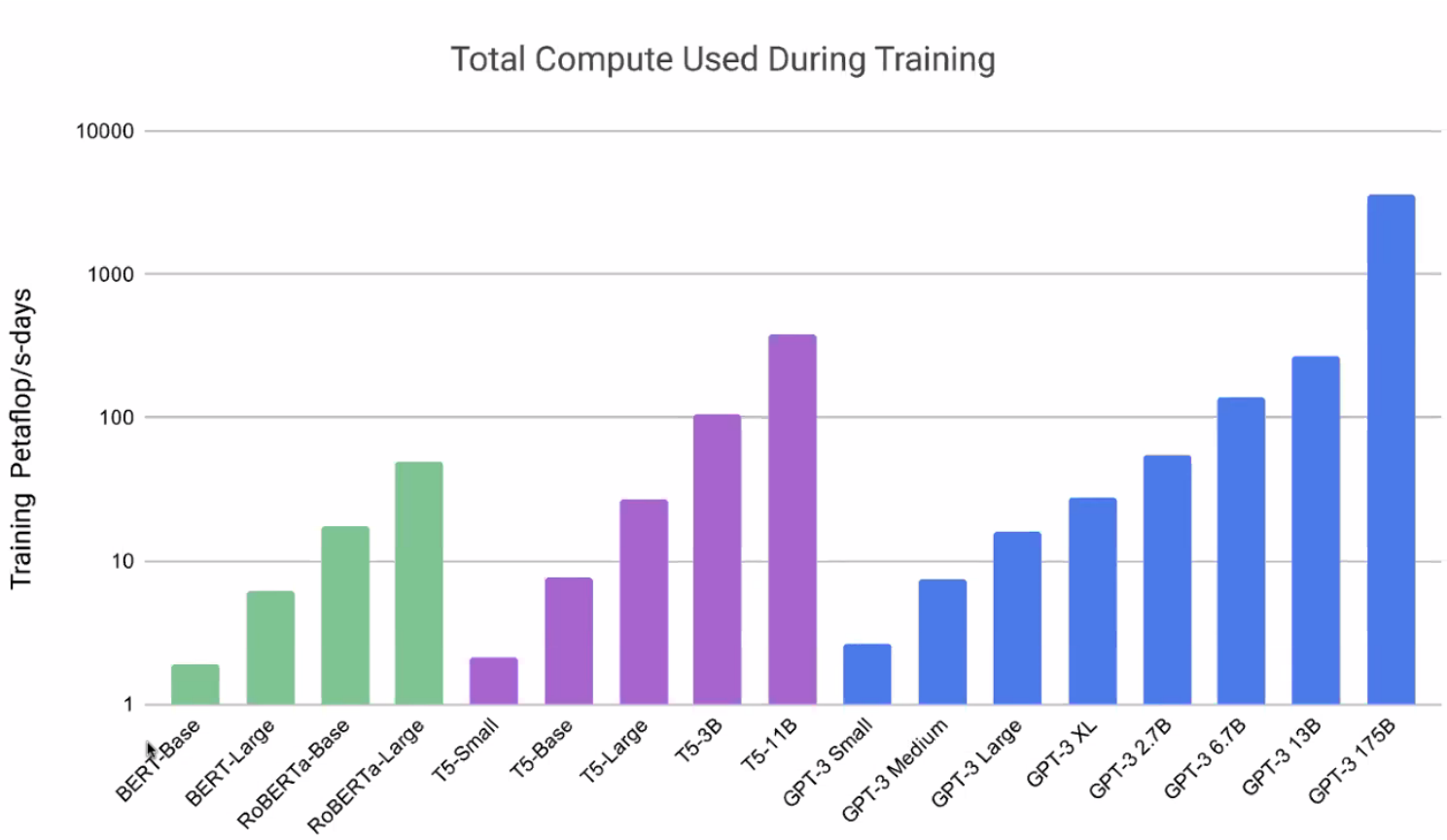
GPT-3, GPT-3.5, InstructGPT, and ChatGPT

- All language models developed by OpenAI
- **GPT-3:** be able to generate human-like text and perform a wide range of natural language processing tasks.
- **GPT-3.5:** is an upgraded version of GPT-3, but with fewer parameters (unknown to public)
- **InstructGPT:** fine-tuned to follow instructions accurately with human feedback to better align with user intentions when providing instructions.
- **ChatGPT** is a model optimized for conversational interactions. It is designed to provide natural and engaging responses in a chat-like format.



LLM example: ChatGPT

- 320 gigabytes of memory that require a minimum of 8 A100 GPUs 40GB





NVIDIA Tesla A100 Ampere 40 GB Graphics Card - PCIe 4.0 - Dual Slot

Brand: Generic

5.0 ★★★★★ 3 ratings

$6,715.00

No Import Fees Deposit & $13.73 Shipping to Hong Kong Details

Eligible for Return, Refund or Replacement within 30 days of receipt

Graphics Coprocessor	NVIDIA Tesla A100
Brand	Generic
Graphics Ram Size	40.00
GPU Clock Speed	1410 MHz
Graphics Processor Manufacturer	NVIDIA

About this item

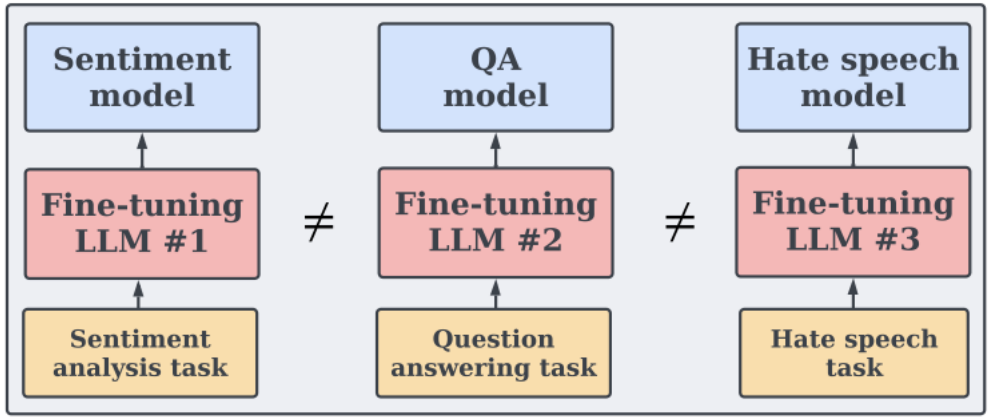
- Standard Memory: 40 GB
- Host Interface: PCI Express 4.0
- Cooler Type: Passive Cooler
- Product Type: Graphics Card

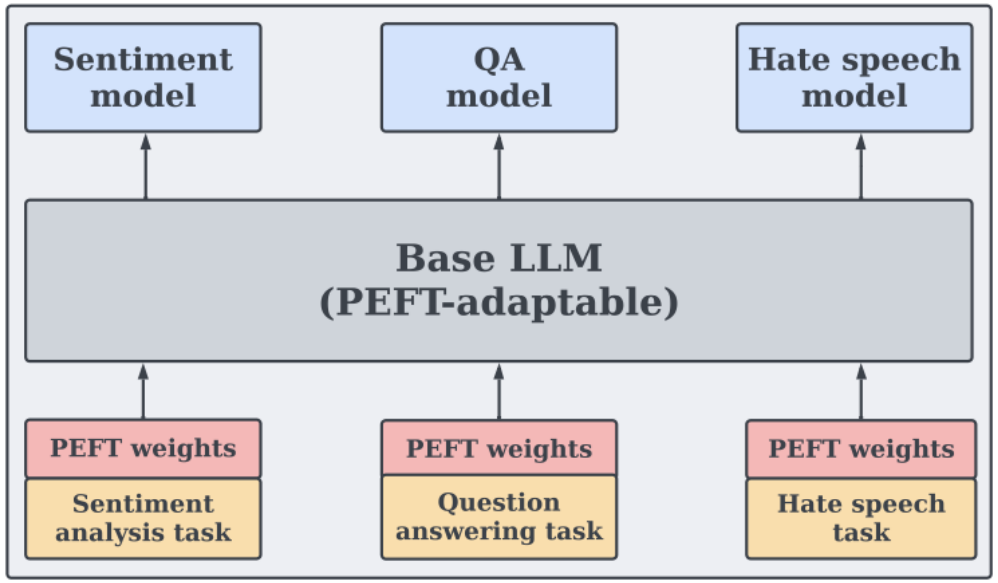
Report incorrect product information.

Note: Products with electrical plugs are designed for use in

Fine-tuning an LLM for a specific down- stream task

- fine-tuning: requires updating the entire model for each task





- LLM: The same base model can be re-used during inference for different tasks

Prompt Engineering

- Primary school

Below is an instruction that describes a task.

Instruction: You are a teacher in a primary school and you would like to plan the topics to be covered in a course. Please draft the 5 topics based on the course title

Course title: What is renewable energy

As a primary school teacher planning a course on "What is renewable energy," here are five potential topics that could be covered:

1. Introduction to Renewable Energy: This topic could cover the basics of renewable energy, including what it is, why it's important, and how it differs from non-renewable sources of energy.
2. Types of Renewable Energy: This topic could explore the various types of renewable energy sources, such as solar, wind, hydro, geothermal, and biomass. Students could learn about the benefits and drawbacks of each type of energy, as well as how they are harnessed.
3. Environmental Impact of Renewable Energy: This topic could examine the positive impact of renewable energy on the environment, including reduced greenhouse gas emissions, improved air and water quality, and reduced dependence on fossil fuels.
4. Renewable Energy in Action: This topic could provide examples of renewable energy in action, such as solar panels on homes or wind turbines on farms. Students could learn about how renewable energy is being used in their community and around the world.
5. The Future of Renewable Energy: This topic could look ahead to the future of renewable energy, including emerging technologies and potential advancements in storage and distribution. Students could explore what the world might look like if we transitioned fully to renewable energy sources.

Prompt Engineering

- kindergarten

Below is an instruction that describes a task.

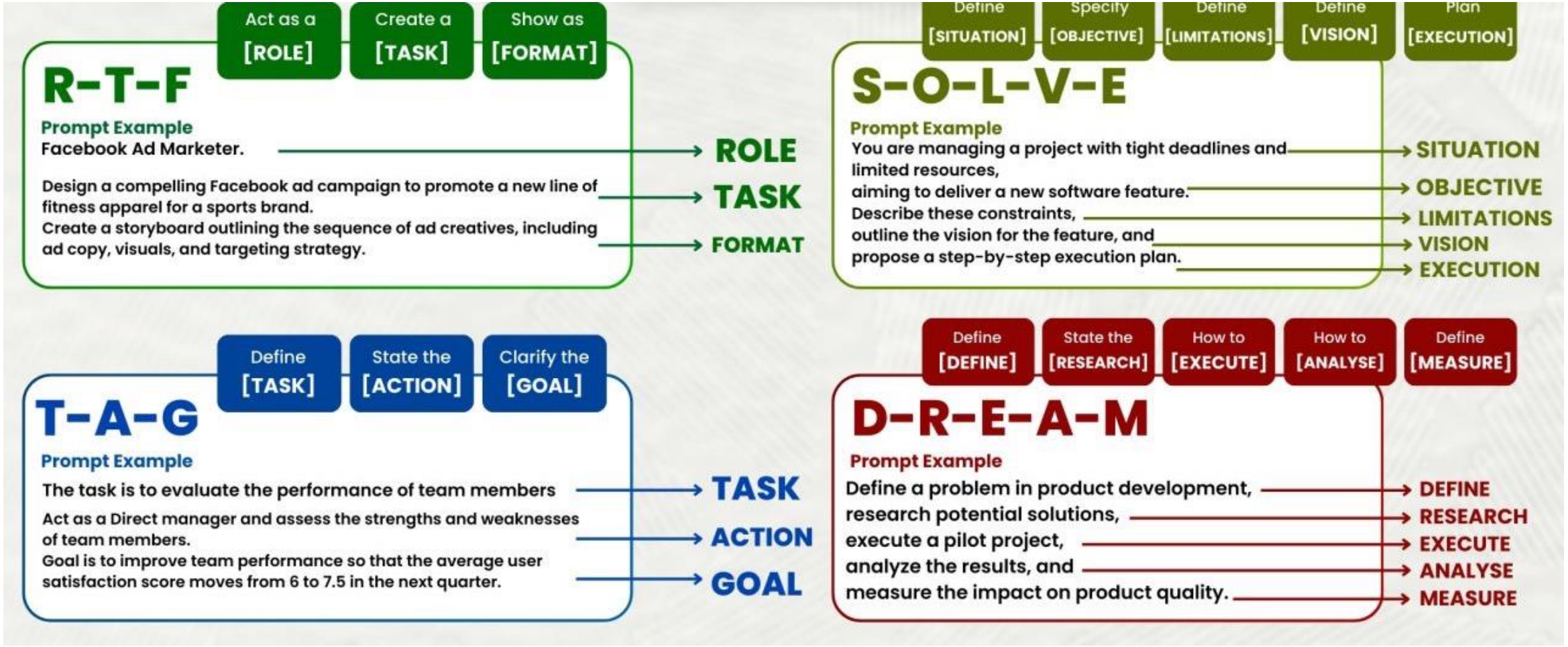
Instruction: You are a teacher in a kindergarten and you would like to plan topics to be covered in a course. Please draft the 5 topics based on the course title

Course title: What is renewable energy

As a kindergarten teacher planning a course on "What is renewable energy," here are five potential topics that could be covered:

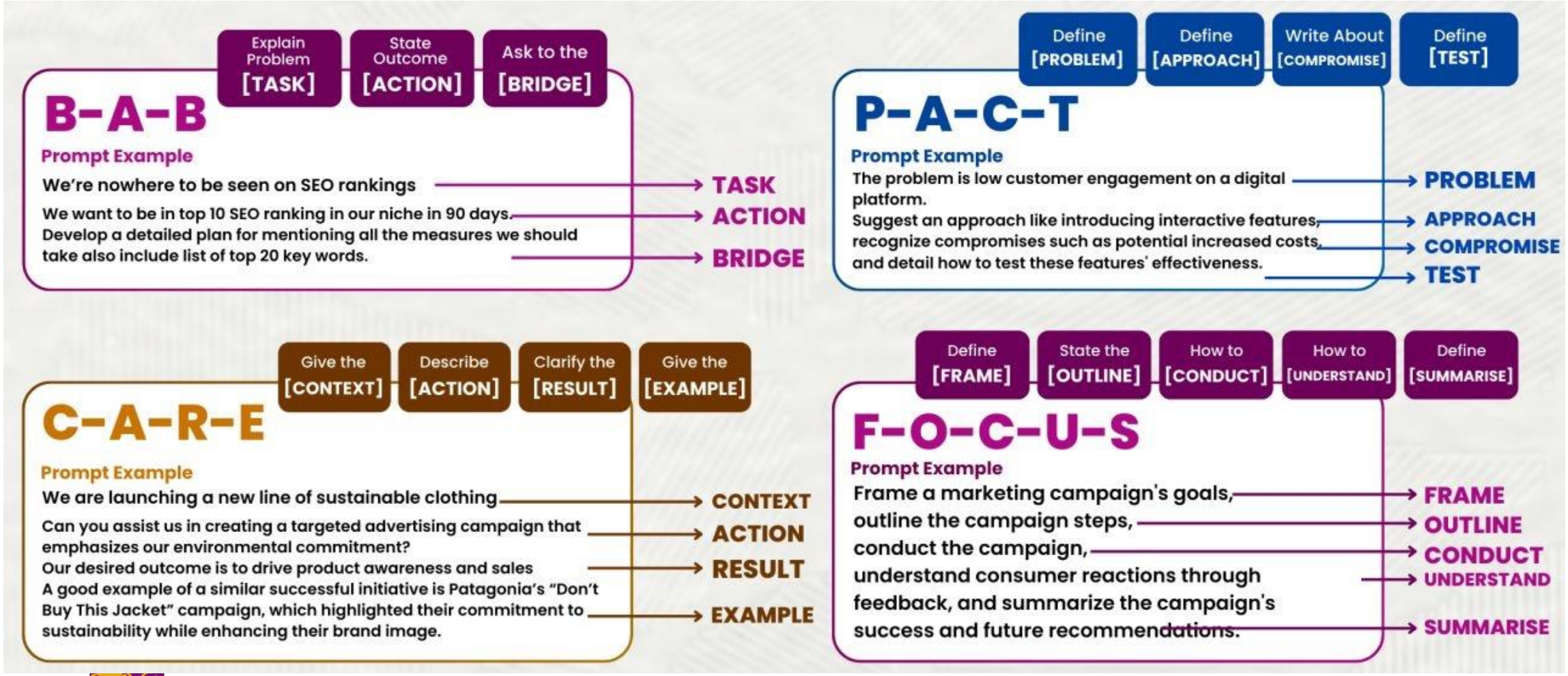
1. What is Energy: This topic could introduce the concept of energy and what it is by using simple examples and demonstrations that are easy for young children to understand.
2. Types of Energy: This topic could introduce the different types of energy, such as light, heat, and motion, and help children understand how energy is used in their everyday lives.
3. Introduction to Renewable Energy: This topic could cover the basics of renewable energy, including what it is, why it's important, and how it differs from non-renewable sources of energy.
4. Types of Renewable Energy: This topic could introduce the various types of renewable energy sources, such as solar, wind, hydro, geothermal, and biomass. Children could learn about how each type of energy is harnessed and how it can be used.
5. Renewable Energy in Action: This topic could provide simple examples of renewable energy in action, such as solar-powered toys or wind turbines. Children could learn how renewable energy is being used in their everyday lives and how it can help protect the environment.

LLM Prompt Frameworks



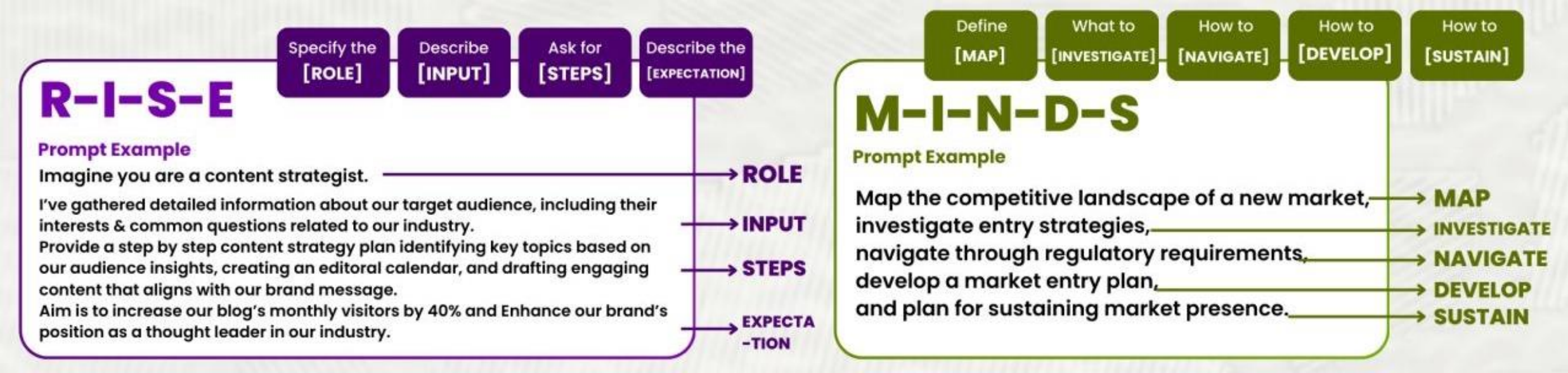
ref: https://www.linkedin.com/posts/denis-panjuta_chatgpt-prompts-frameworks-key-frameworks-activity-7201492501529079810-B1x7/

LLM Prompt Frameworks



ref: https://www.linkedin.com/posts/denis-panjuta_chatgpt-prompts-frameworks-key-frameworks-activity-7201492501529079810-B1x7/

LLM Prompt Frameworks



ref: https://www.linkedin.com/posts/denis-panjuta_chatgpt-prompts-frameworks-key-frameworks-activity-7201492501529079810-B1x7/

Zero-shot chain-of-thought prompting (More later)

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

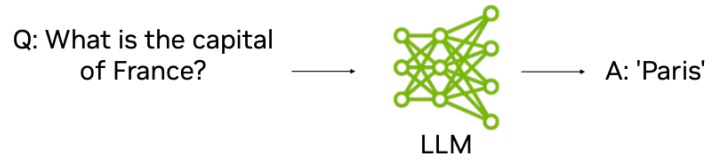


Ref: **Large Language Models are Zero-Shot Reasoners**

Zero-shot vs. Few-shot (More later)

Zero-Shot

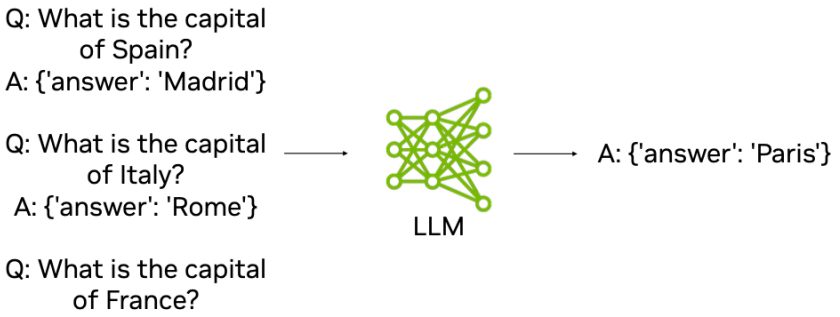
Asking the foundation model to perform a task with no in-prompt example



Lower token count
More space for context

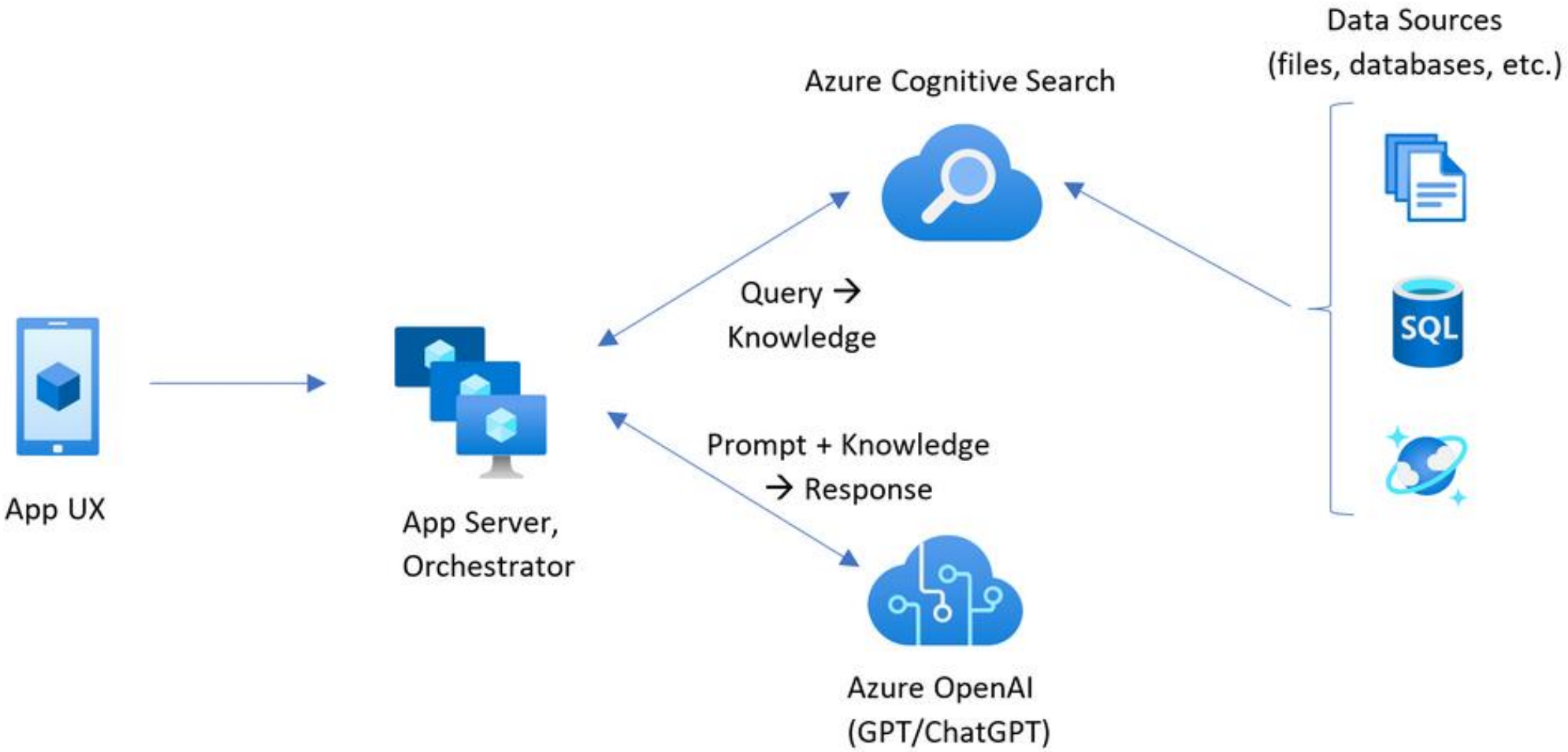
Few-Shot

Providing examples as context to the foundation model related to a task

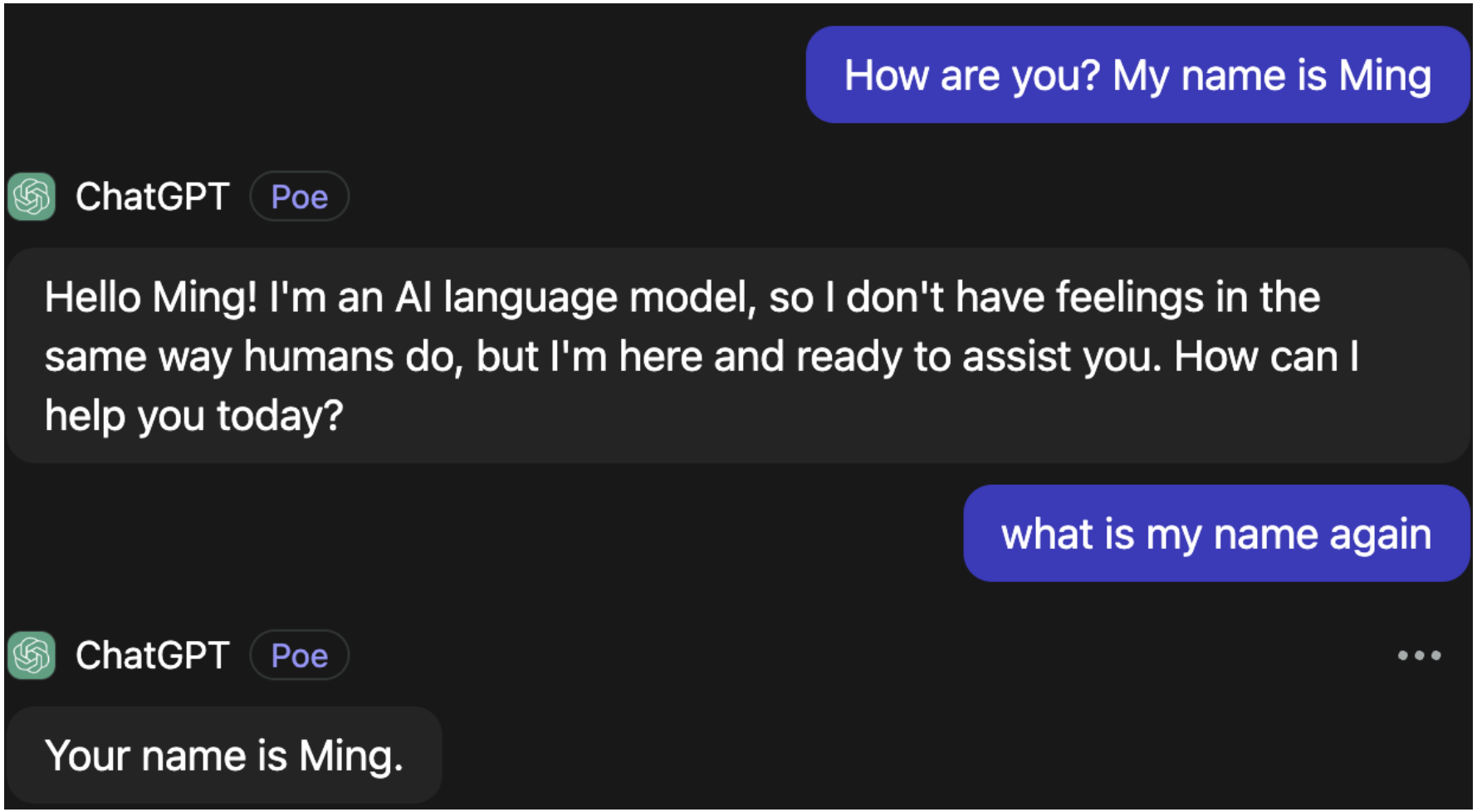


Better aligned responses
Higher accuracy on complex questions

ChatGPT on Azure



How does LLM “remember” the Conversation



How does LLM “remember” the Conversation - Demo

```python
previous = []

temp = {}
temp["role"] = 'user'
temp["content"] = "How are you? I am Ming"
previous.append(temp)

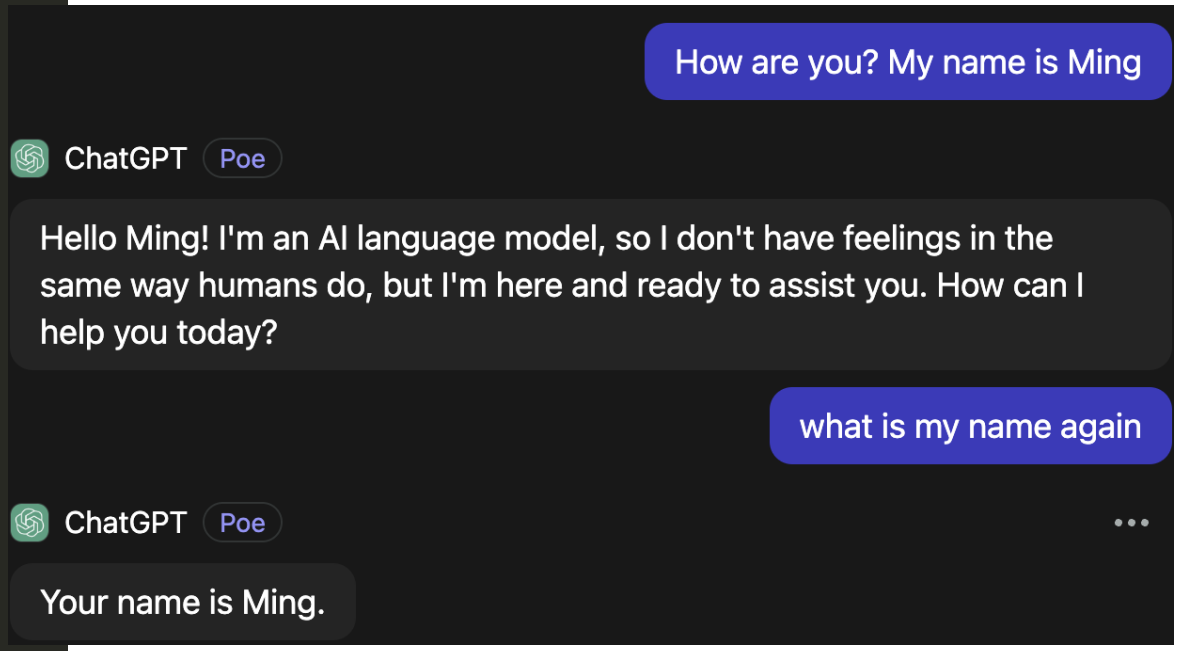
response_message = chgatbot.call(previous)

temp = {}
temp["role"] = 'system'
temp["content"] = response_message
previous.append(temp)

temp = {}
temp["role"] = 'user'
temp["content"] = "What is my name again"
previous.append(temp)

response_message = chgatbot.call(previous)

```



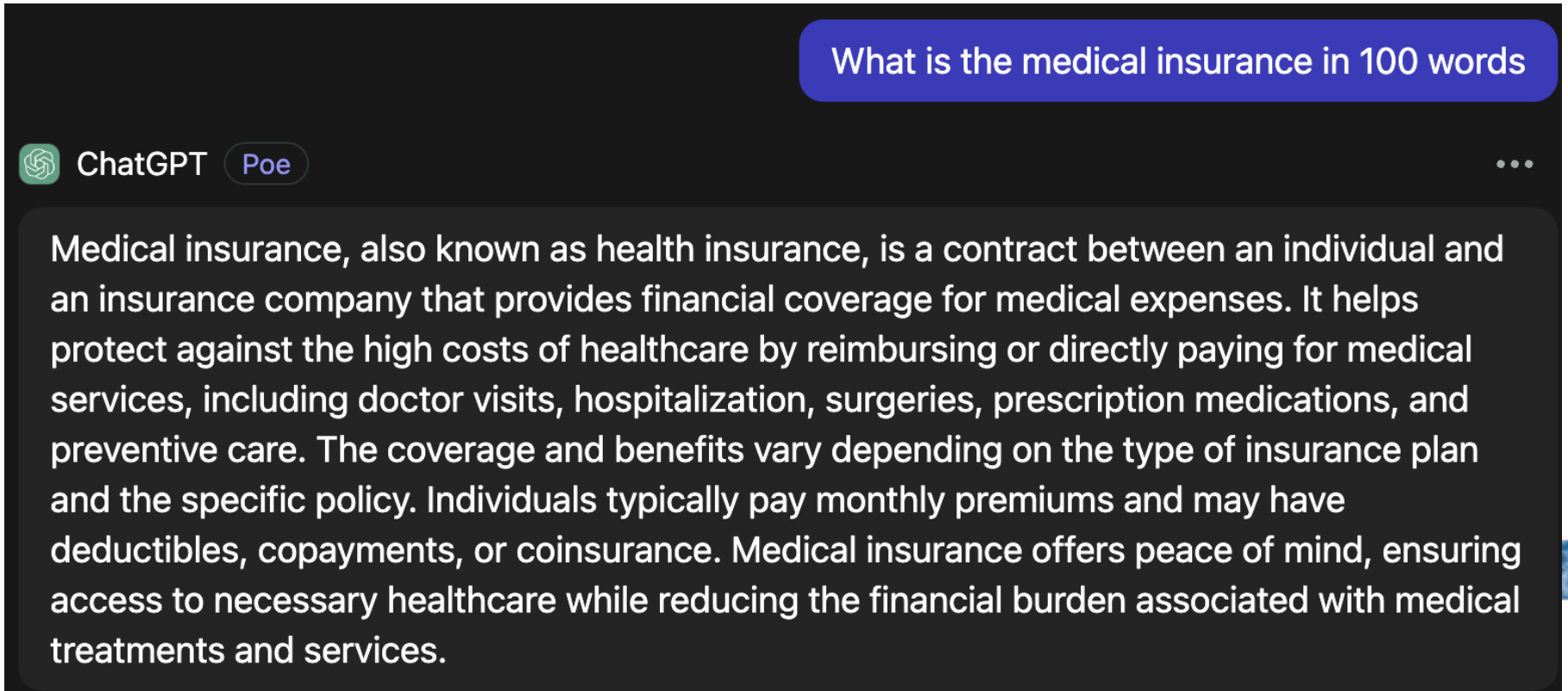
Agenda

- What is NLP and Large Language Model (LLM)
- Retrieval Augmented Generation (RAG)
- Agentic AI



Retrieval Augmented Generation (RAG)

- Motivations
  - Decouples an LLM from only being able to act on original training data
  - Obviates the need to retrain the LLM with the latest data
  - LLMs limited by context window sizes



Retrieval Augmented Generation (RAG)

- Concepts
  - Connect LLM to data sources at inference time
  - e.g., Databases, Web, Documents, 3rd Party APIs, etc.
  - Find relevant data
  - Inject relevant data into the prompt

Medical insurance is a type of coverage that provides financial protection for medical expenses. It is designed to help individuals and their families afford healthcare services and treatments. With a minimum of 10 employees, you can create a plan that includes basic benefits such as hospital room and board, surgeon's fee, and specialist's fee. You also have the option to add supplementary major medical benefits for additional coverage for serious illnesses and injuries beyond basic hospitalization. Optional benefits like outpatient care, dental care, and maternity benefits can be included as well. You can customize the plan based on different employee categories and their dependents. One advantage is the waiver of medical underwriting and health declaration, making it easier to obtain coverage.



Small is beautiful

You can set up a plan with as few as 10 employees.



Design your own plan

Customise your plan with different levels of Hospitalisation, Supplementary Major Medical, Outpatient, Dental and/or Maternity Benefits for different categories of employees and their dependants*.



Basic benefits

Basic coverage includes but not limited to hospital room & board, surgeon's fee, anaesthetist's fee, operating theatre fee, specialist's fee under Hospitalisation Benefits.



Waiver of underwriting

We waive medical underwriting and health declaration.



Optional benefits

In addition to basic Hospitalisation Benefits, you can opt for Supplementary Major Medical Benefits for extra protection for serious illnesses and injuries in excess of basic hospitalisation coverage. You can also opt for Outpatient Benefits, Dental Benefits and/or Maternity Benefits.

https://www.fwd.com.hk/en/business/employee-benefits/caring-employee-benefit-insurance-plan/

RAG for AI

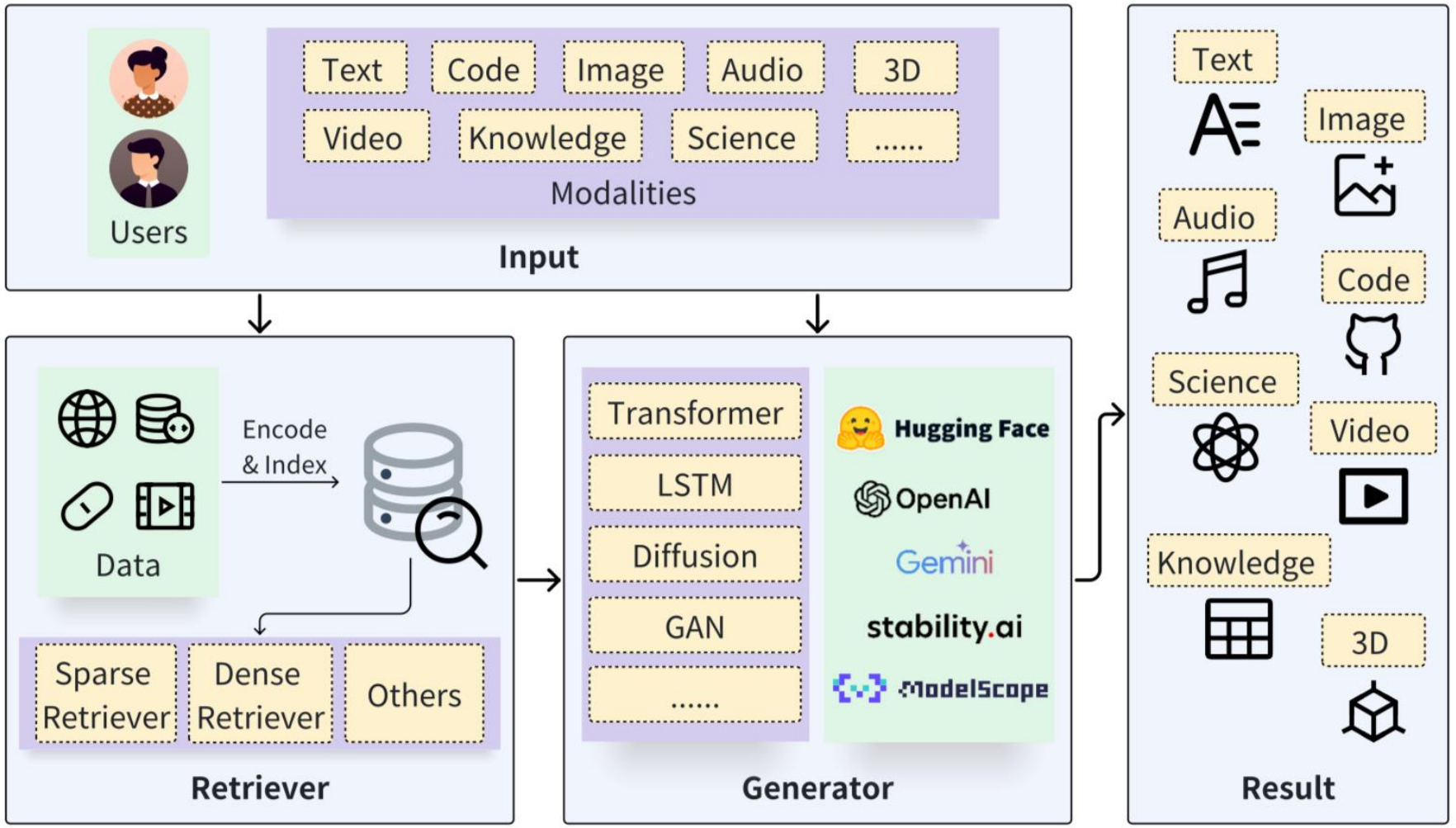
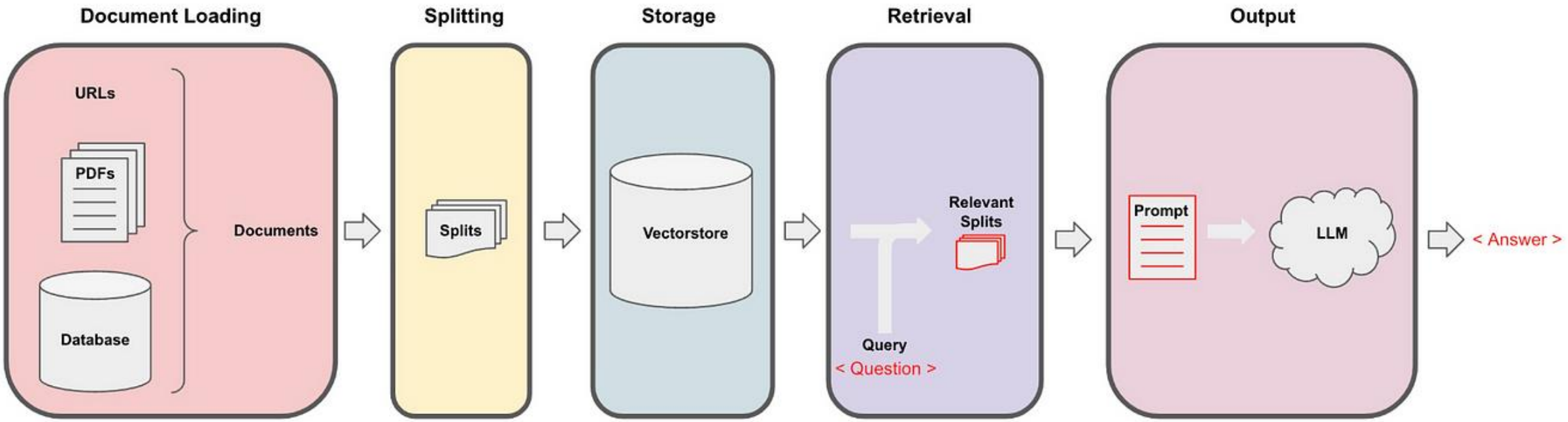


Fig. 1. A generic RAG architecture. The users' queries, which may be different modality, serve as input to both the retriever and the generator. The retriever searches for relevant data sources in storage, while the generator interacts with the retrieval results and ultimately produces results of various modalities.

Ref: Retrieval-Augmented Generation for AI-Generated Content: A Survey

Document Processing



Simple Local Vector DB

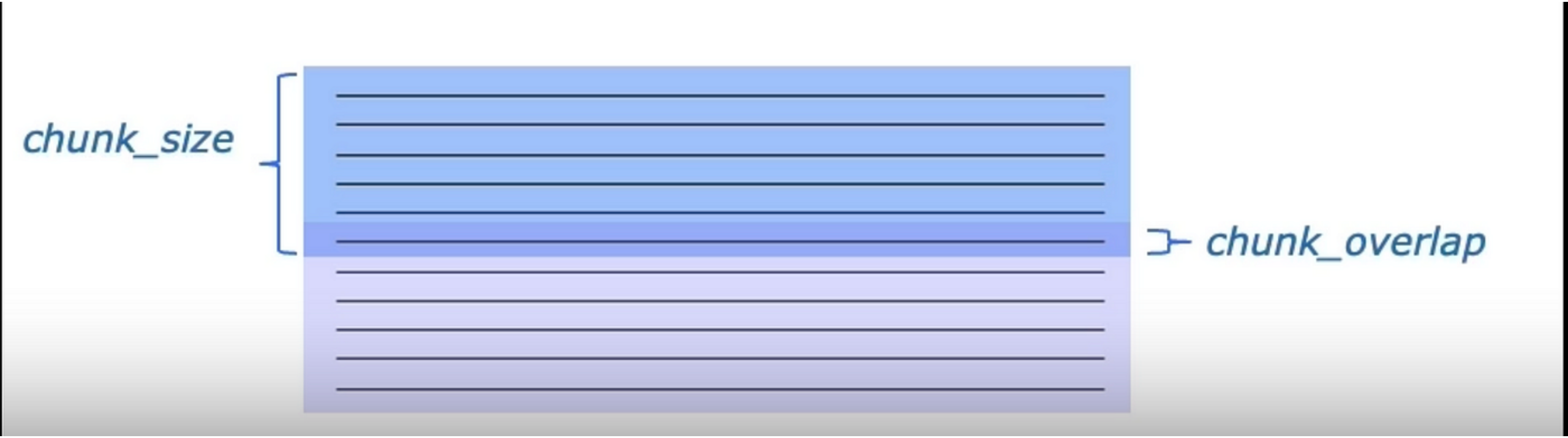


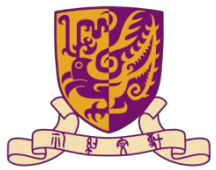
Splitting algorithms

- **CharacterTextSplitter-** Implementation of splitting text that looks at characters.
- **MarkdownHeaderTextSplitter** - Implementation of splitting markdown files based on specified headers.
- **TokenTextSplitter** - Implementation of splitting text that looks at tokens.
- **SentenceTransformersTokenTextSplitter** - Implementation of splitting text that looks at tokens.
- **RecursiveCharacterTextSplitter** - Implementation of splitting text that looks at characters. Recursively tries to split by different characters to find one that works.
  Language) - for CPP, Python, Ruby, Markdown etc
- **NLTKTextSplitter** - Implementation of splitting text that looks at sentences using NLTK (Natural Language Tool Kit)
- **SpacyTextSplitter** - Implementation of splitting text that looks at sentences using Spacy



RecursiveSplitting





RecursiveSplitting

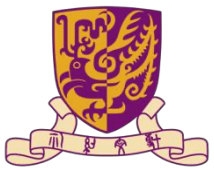
```
some_text = """When writing documents, writers will use document structure to group content. \
This can convey to the reader, which idea's are related. For example, closely related ideas \
are in sentances. Similar ideas are in paragraphs. Paragraphs form a document. \n\n  \
Paragraphs are often delimited with a carriage return or two carriage returns. \
Carriage returns are the "backslash n" you see embedded in this string. \
Sentences have a period at the end, but also, have a space.\
and words are separated by space."""

len(some_text) -> 496

c_splitter = CharacterTextSplitter(
    chunk_size=450,
    chunk_overlap=0,
    separator = ' '
)
r_splitter = RecursiveCharacterTextSplitter(
    chunk_size=450,
    chunk_overlap=0,
    separators=["\n\n", "\n", " ", ""]
)
```

```
['When writing documents, writers will use document structure to group content.
 'have a space.and words are separated by space.']
```

```
['When writing documents, writers will use document structure to group content.
 'Paragraphs are often delimited with a carriage return or two carriage returns.
```



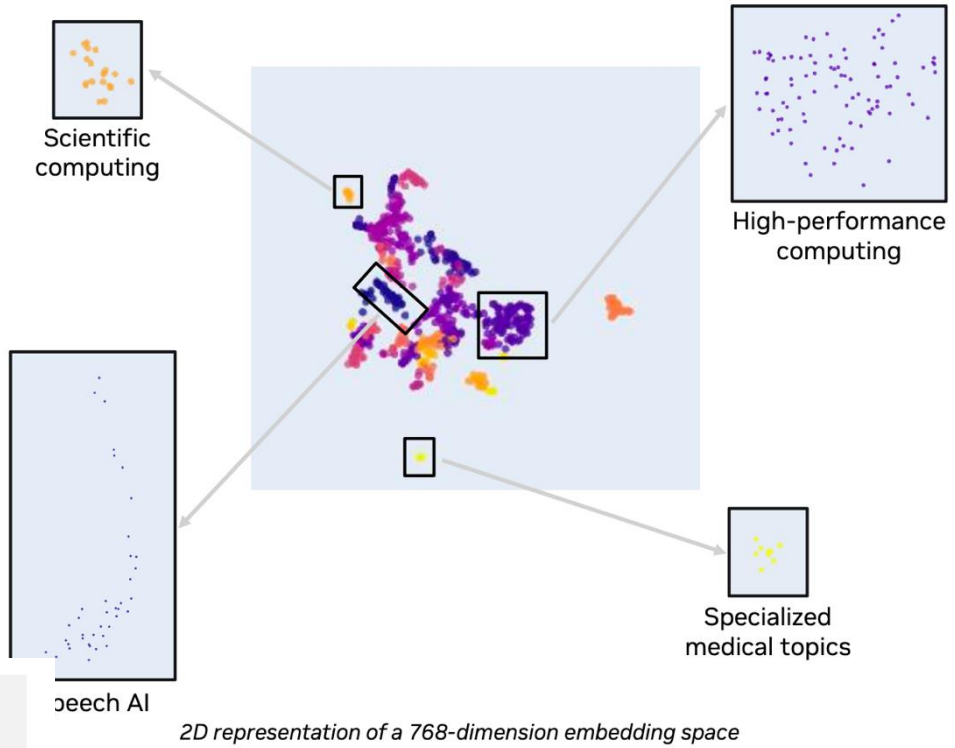
Fast vs slow retrievers

- Fastest: computing similarities (or word overlap) on pre-computed vectors
- Slowest: using a LM to compute similarities
- Other hybrid (pre-computing the ‘document index’)



Similarity search

- Embeddings are data (text, image, or other data) represented as numerical vectors
- Trained models to embed input but keywords only

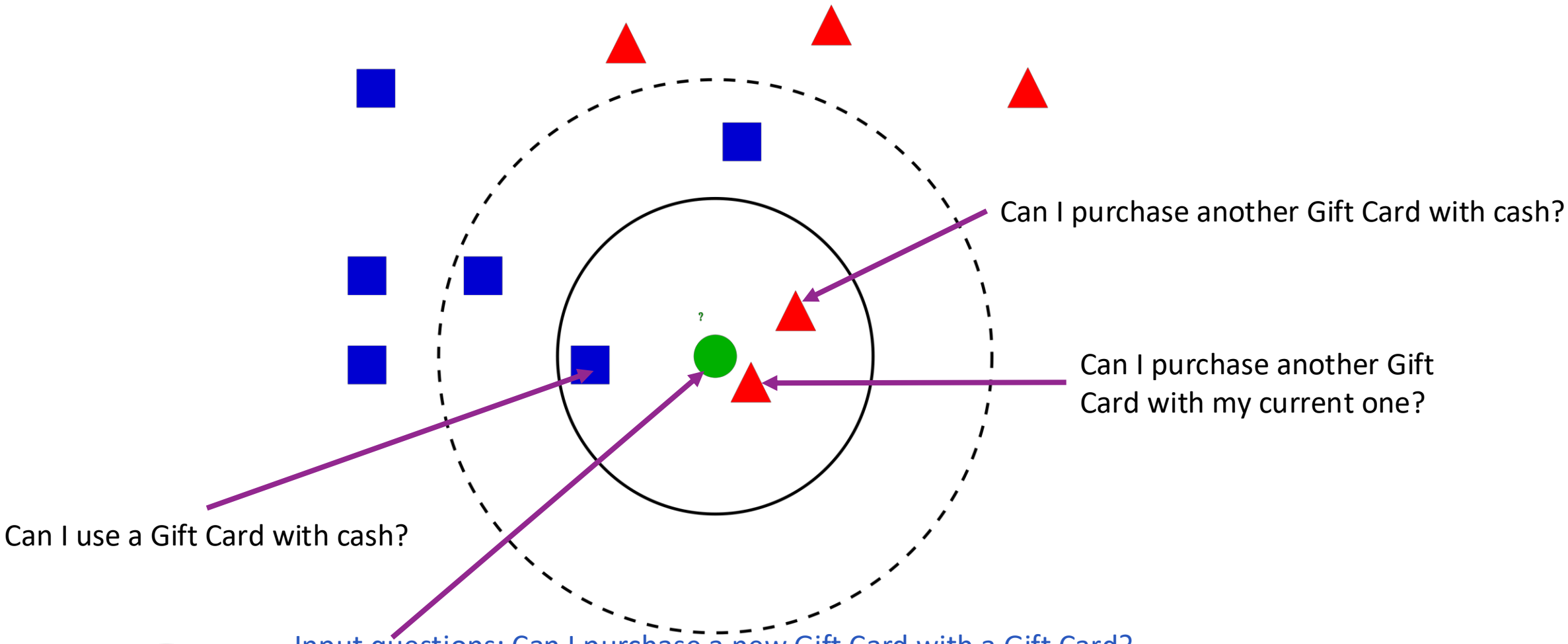


2D representation of a 768-dimension embedding space

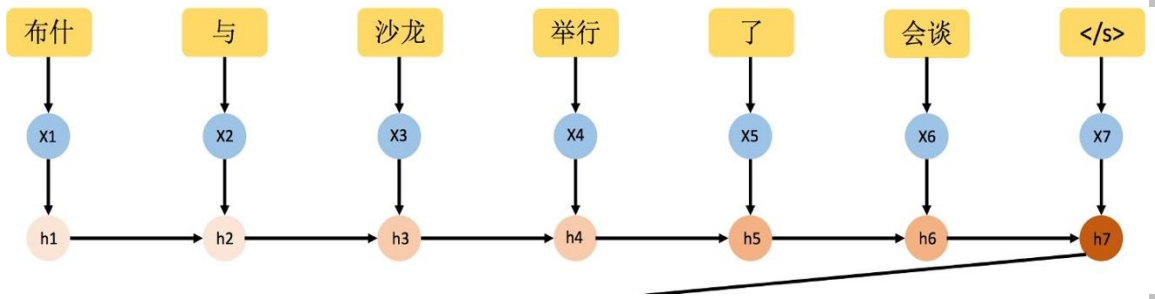
Query: Who will lead the construction team?
Chunk 1: The construction team found lead in the paint.
Chunk 2: Ozzy has been picked to lead the group.

Chunk 1 shares more keywords with the query, but semantic search can differentiate the meanings of "lead" and understand that "team" and "group" are similar, so Chunk 2 may be more helpful for the query.

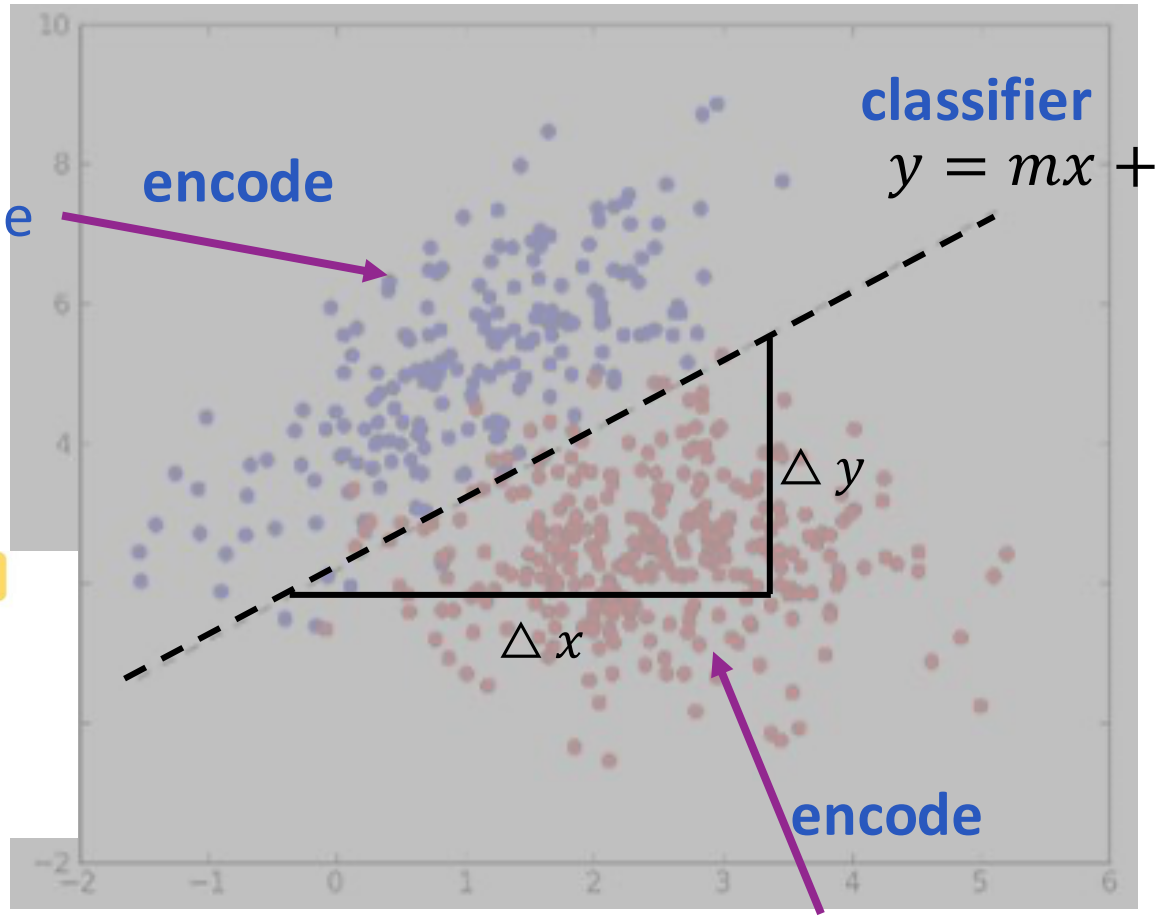
Flow of Data Augmentation for RAG



This is a good movie
I highly recommend this movie
I will watch it again and again



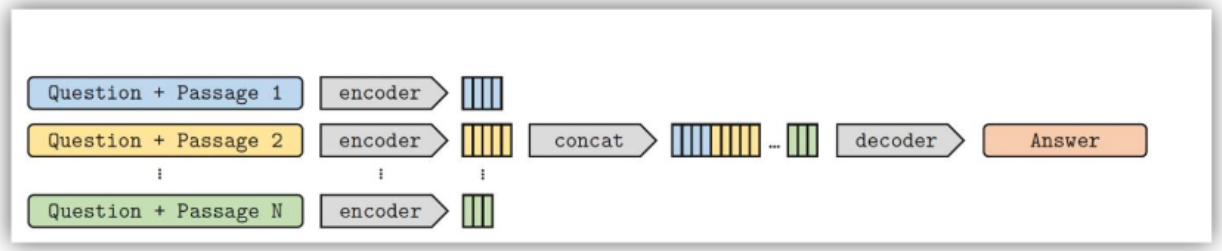
Pre-train model (encoder)



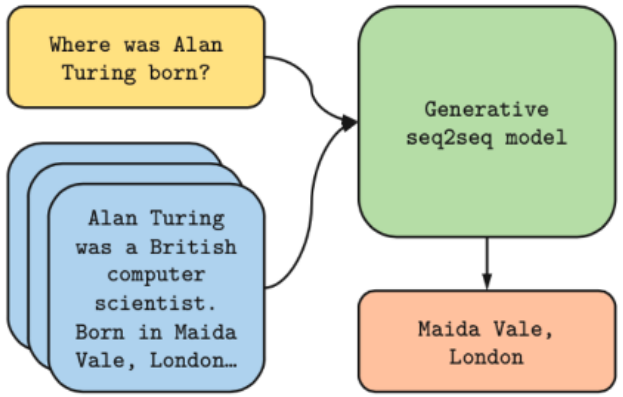
This is not a good movie
I won't recommend this movie
I will not watch it again

Deep retrieval + generative models

- Recent work shows that it is beneficial to generate answers instead of to extract answers.



Fusion-in-decoder (FID)=DPR+T5



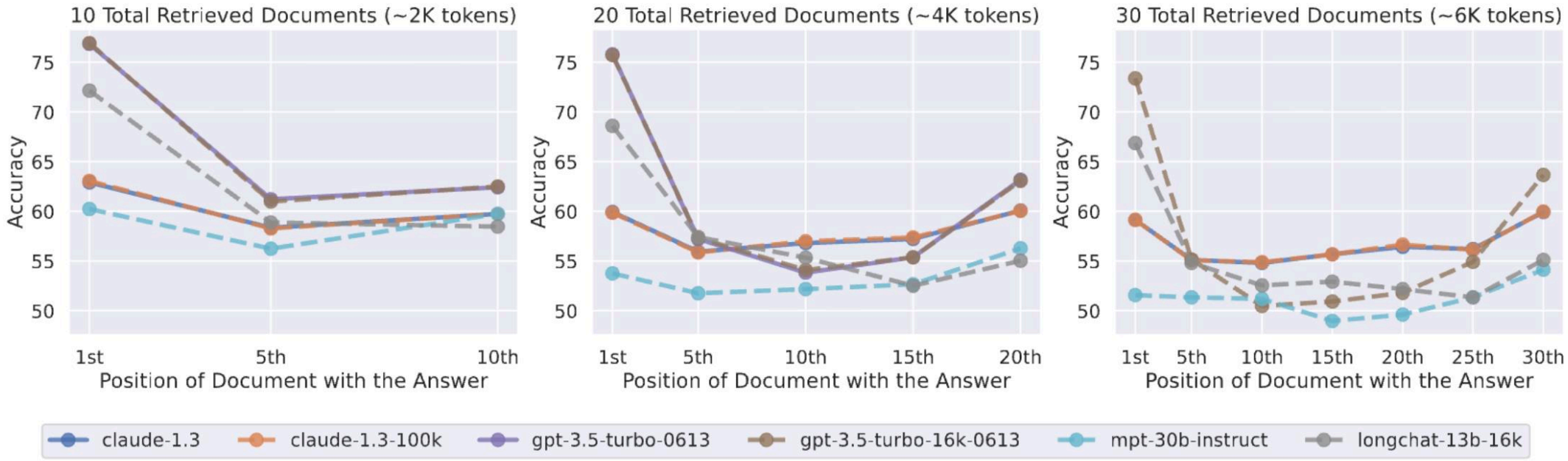
Model	NaturalQuestions	TriviaQA	
ORQA (Lee et al., 2019)	31.3	45.1	-
REALM (Guu et al., 2020)	38.2	-	-
DPR (Karpukhin et al., 2020)	41.5	57.9	-
SpanSeqGen (Min et al., 2020)	42.5	-	-
RAG (Lewis et al., 2020)	44.5	56.1	68.0
T5 (Roberts et al., 2020)	36.6	-	60.5
GPT-3 few shot (Brown et al., 2020)	29.9	-	71.2
Fusion-in-Decoder (base)	48.2	65.0	77.1
Fusion-in-Decoder (large)	**51.4**	**67.6**	**80.1**

Izacard and Grave 2020. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering



LLM can't pay attention to the entire context

- The long-context problem bites you – LMs do not pay attention to its context well!
- Setup: 1 relevant document, all others irrelevant



SpaCy

Get things done

spaCy is designed to help you do real work — to build real products, or gather real insights. The library respects your time, and tries to avoid wasting it. It's easy to install, and its API is simple and productive.

GET STARTED

Blazing fast

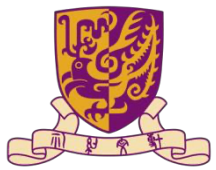
spaCy excels at large-scale information extraction tasks. It's written from the ground up in carefully memory-managed Cython. If your application needs to process entire web dumps, spaCy is the library you want to be using.

FACTS & FIGURES

Awesome ecosystem

In the five years since its release, spaCy has become an industry standard with a huge ecosystem. Choose from a variety of plugins, integrate with your machine learning stack and build custom components and workflows.

READ MORE



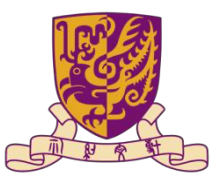
Ref: https://spacy.io/

en_core_web_sm – training dataset

- OntoNotes - a large corpus of English news articles and transcripts of spoken language, annotated with part-of-speech tags, named entities, and syntactic dependencies.
- WikiNER - a corpus of English Wikipedia articles, annotated with named entities.
- WebText - a collection of web pages, primarily from educational domains, that are suitable for language modeling and text classification.
- English Gigaword - a large corpus of English news articles, annotated with named entities.

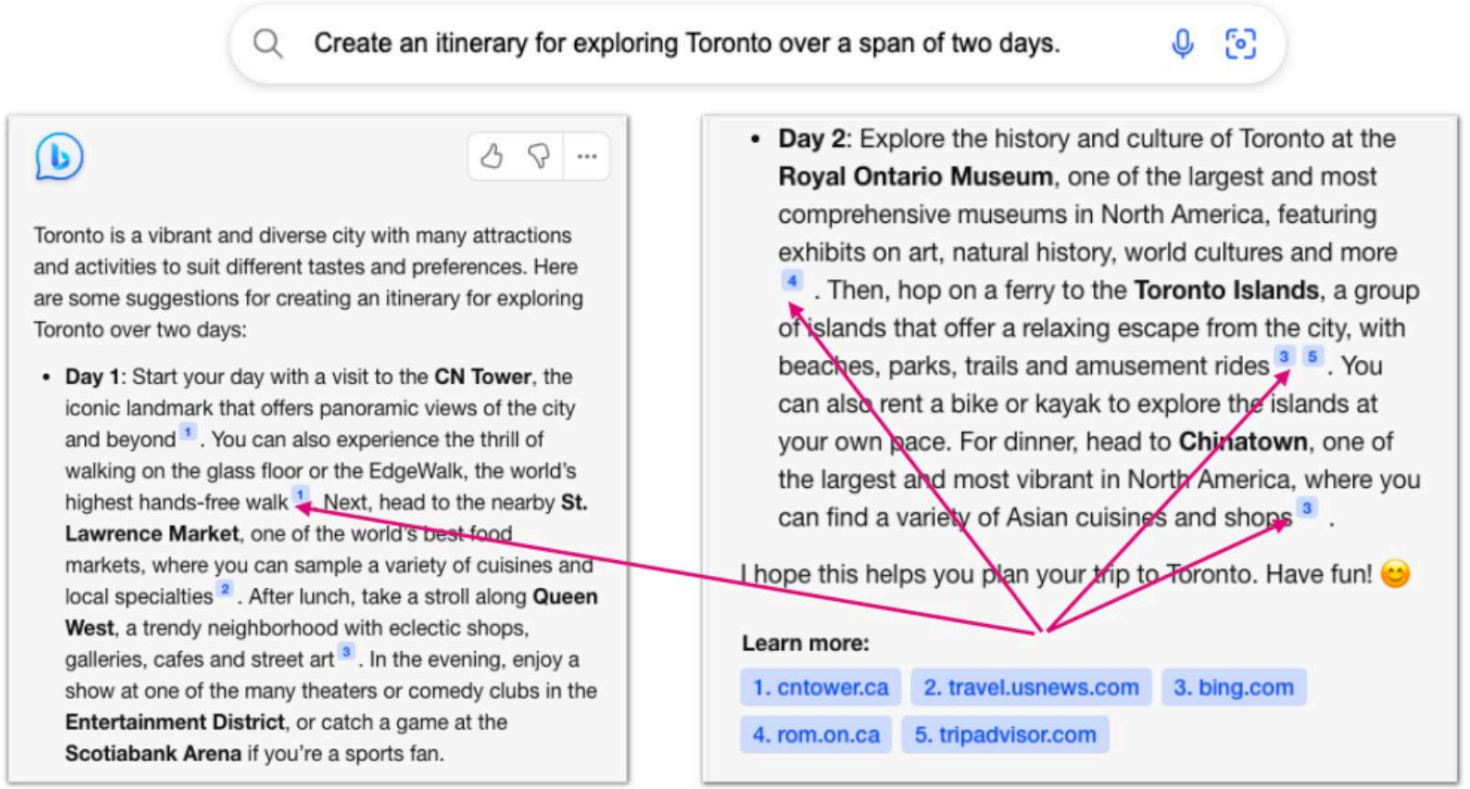


Search Method	Strengths	Weaknesses	Best For
Vector Search	• Excellent semantic understanding • Finds contextually relevant information • Works well with paraphrased content	• Poor with uncommon abbreviations • Misses exact matches of specific terms • Struggles with organization-specific terminology	Finding semantically similar content that may use different terminology
BM25 (Keyword)	• Excellent at exact matches • Computationally efficient • Handles specific terminology well	• Limited semantic understanding • Misses relevant content with different wording • Ignores document context	Finding documents containing specific words, names, or abbreviations
Hybrid + Reranking	• Combines strengths of both approaches • Adjustable weighting between methods • Reranker improves relevance scoring	• More complex to implement • Requires tuning of weight parameter • Potentially higher computational cost	Comprehensive search systems needing both semantic understanding and exact matching



Note: LLMs citation and hallucinations

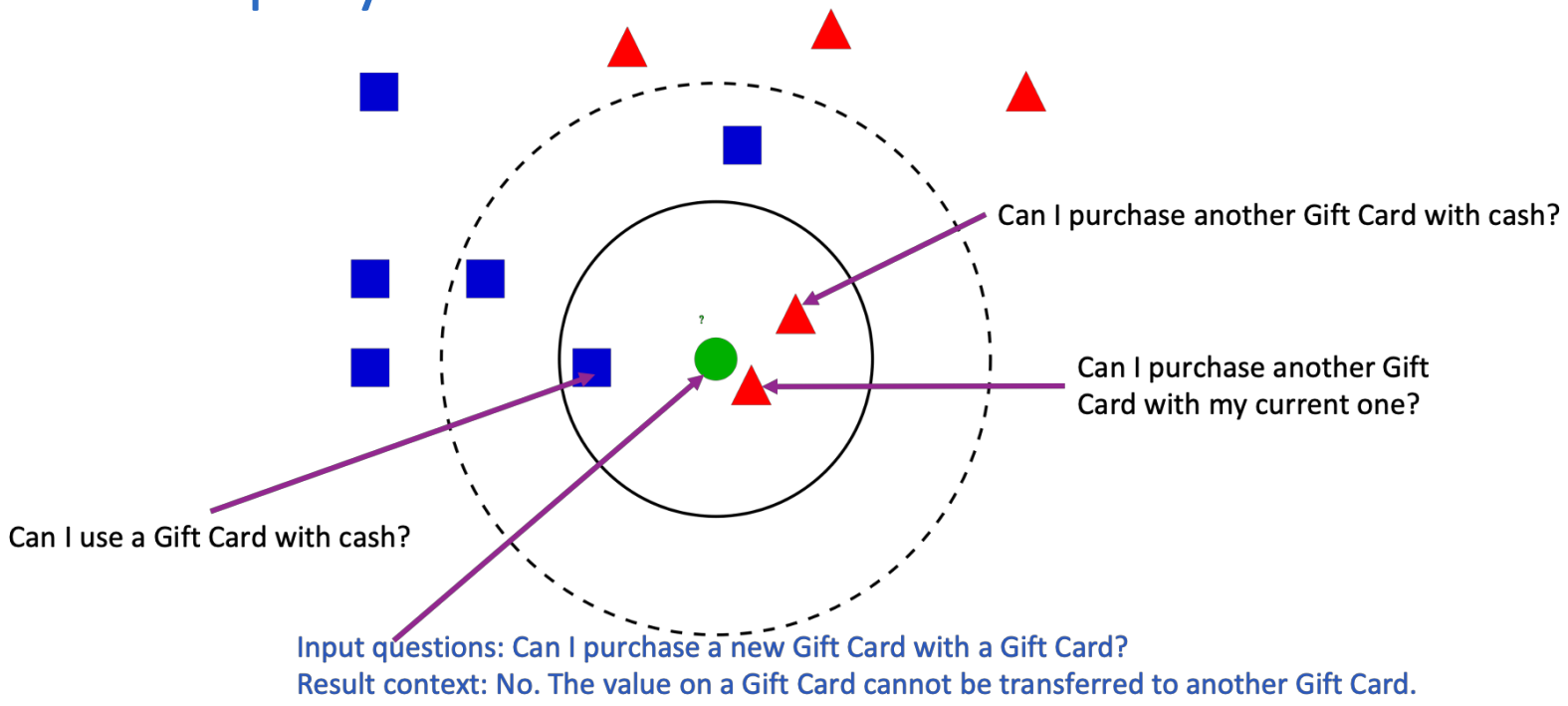
- Unique benefit of RAG: citing the source
- These citations themselves are generated by LLMs
- The citations could also be hallucinated



Demo

- Extract texts from PDF, Encode the chunks using Spacy

- Compare similarity of text with Spacy to find the most similar one



Grok

cheap EV in HK

Searched for "site:gov.hk electric vehicle incentives Hong Kong 2025" 10 results

- Promotion of Electric Vehicles | Environmental Protection Department epd.gov.hk
- Transport Department - First Registration Tax Concessions for Electric Vehicles td.gov.hk
- The EV-charging at Home Subsidy Scheme evhomecharging.gov.hk
- Extension of first registration tax concession arrangement for electric vehicles for two years and re... info.gov.hk
- Government announces Hong Kong Roadmap on Popularisation of Electric Vehicles - Carbon Neu... cnsd.gov.hk
- Applications for Trial | Environment and Ecology Bureau eeb.gov.hk

Searched for "affordable EVs Hong Kong prices September 2025" 15 results

- [Continuously updated] Hong Kong's latest electric car recommendations for 2025 | One-for-on... kwiksure.com
- Electric Car Hong Kong Price Comparison: Tesla, Mercedes-Benz, KIA, MG, BMW | Money... blog.moneysmart.hk
- The Electric Car Market (EV'S) in Hong Kong 2025 — J3 Private Tours | J3 Consulta... j3consultantshongkong.com
- Ultimate Guide To Buying An Electric Car In Hong Kong 2025 - Little Steps littlestepsasia.com
- Latest Electric Car Listings and Price in HK - BuyCar.hk buycar.hk
- Tesla Car Prices In Hong Kong - New Tesla Car Models 2025 - 2026 - CCP HKG ccarprice.com
- Top 10 Electric Car Model in Hong Kong – April 2025 - Navigator Insurance navigator-insurance.com
- Price List | Hyundai Hong Kong hyundai.com

Perplexity

can you introduce Memory in agentic AI? Such as, Compression & Consolidation Memory. Find some related research papers

Perplexity | Steps | Sources · 92

growthjockey.com
What is Agentic AI Memory? Importance &...

ibm.com
What Is AI Agent Memory? | IBM

DiamantAI
Memory Optimization Strategies in AI Agents -...

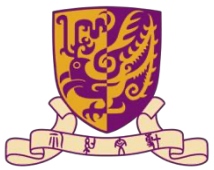
Brain Sciences
A Hippocampus-Inspired Approach to the Stability–...

Memory in Agentic AI: Compression & Consolidation

Introduction

Memory is the cornerstone of intelligent behavior in both biological and artificial systems. In agentic AI, memory mechanisms enable autonomous agents to learn from past experiences, maintain context across interactions, and adapt to changing environments without catastrophic forgetting. Unlike traditional AI models that process each interaction independently, agentic AI systems with sophisticated memory architectures can remember past states, accumulate knowledge over time, and provide personalized, context-aware responses. growthjockey +2

The integration of memory systems in agentic AI draws heavily from neuroscience research, particularly insights from hippocampal-cortical memory consolidation processes. Two critical mechanisms have emerged as fundamental to building robust memory systems: **compression**



Agenda

- What is NLP and Large Language Model (LLM)
- Retrieval Augmented Generation (RAG)
- Agentic AI

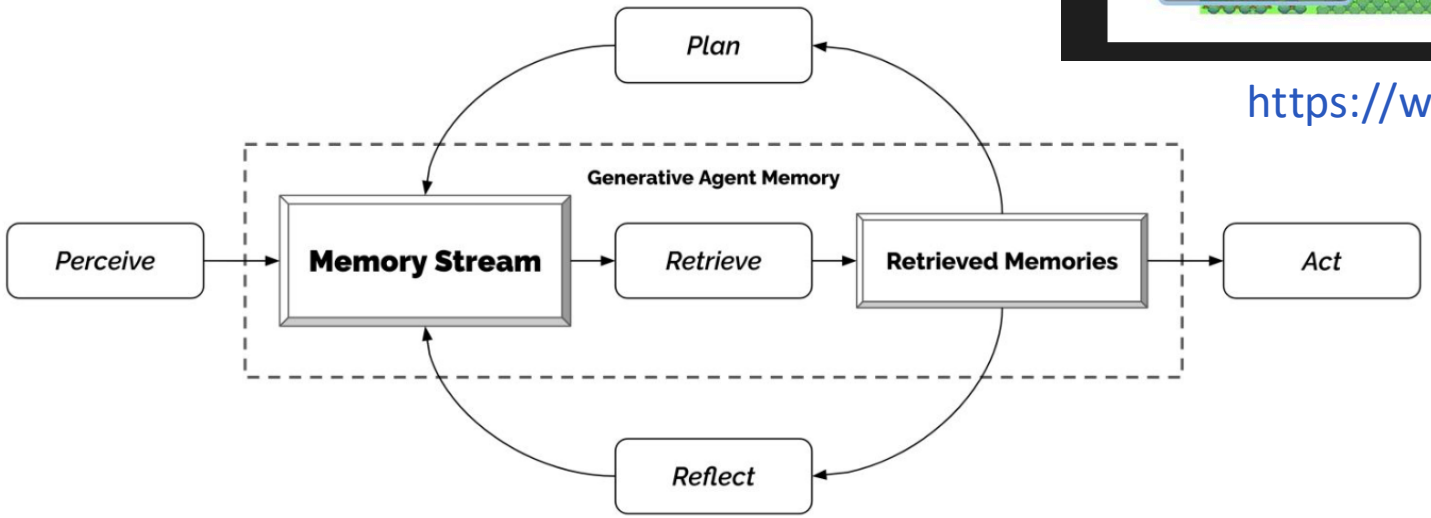


Generative Agents

- 25 Agents
- The Smallville sandbox world
- Emergent Social Behaviors
- With memory and observations

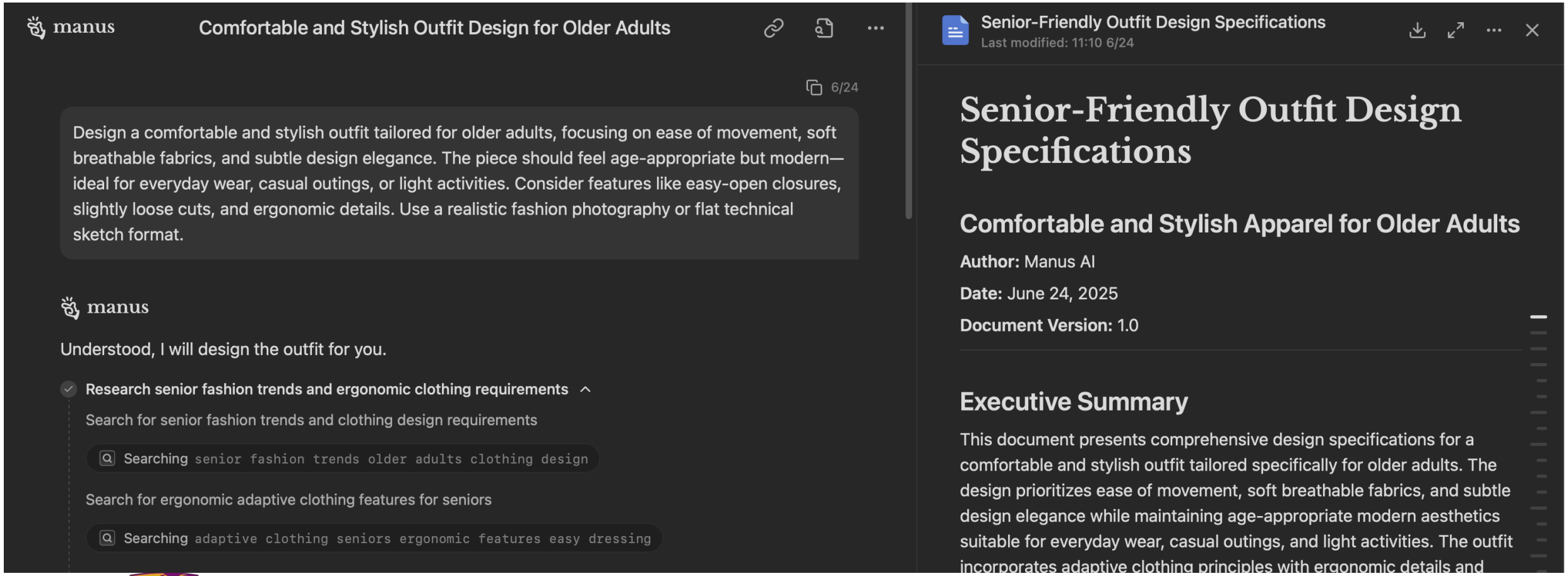


https://www.youtube.com/watch?v=ZdoU9vI2yCg



Manus

manus



https://manus.im/share/8d4HW181SMU0Xf56ylt2zj?replay=1

Auto Data Scientist

- Development Stage: Utilizes CBR to create an iterative pipeline that improves model performance based on historical data.
- Deployment Stage: Adapts successful solutions from the development stage for direct code generation, minimizing the resource demands on LLMs.

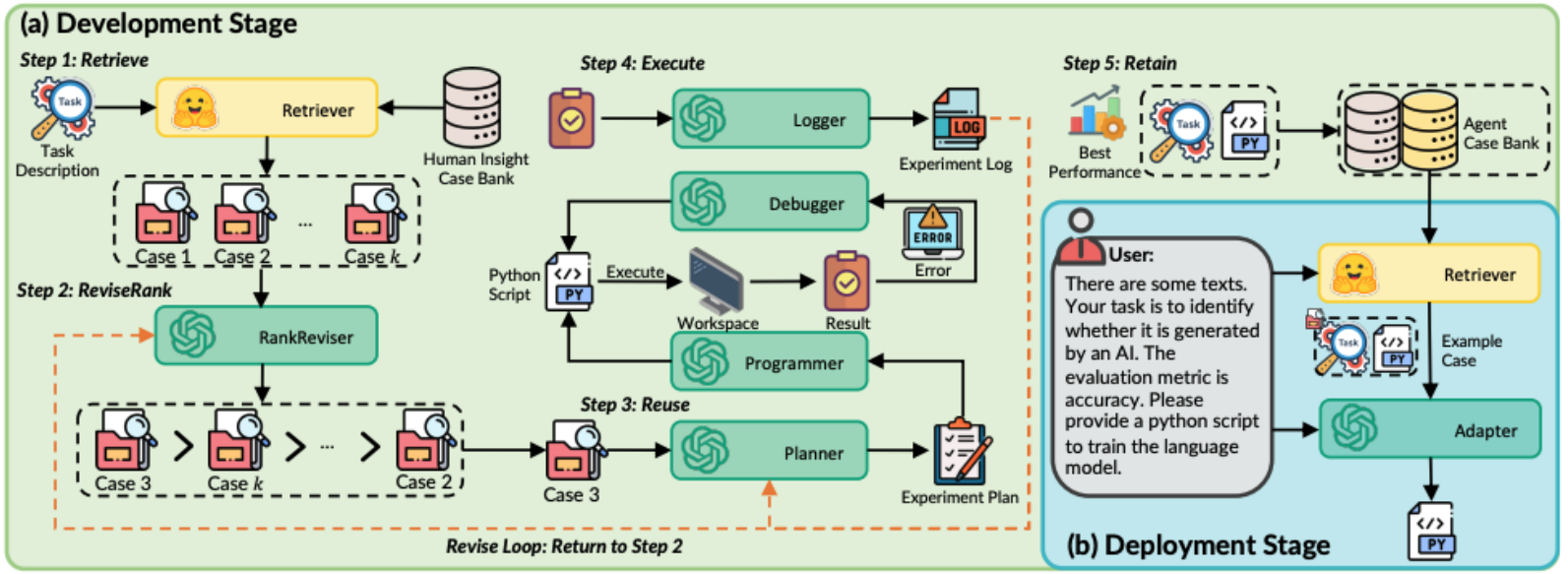


Figure 3. The diagram of DS-Agent. **(a) Development Stage:** DS-Agent structures an automatic iteration pipeline to build and revise the model based on execution feedback. **(b) Deployment Stage:** DS-Agent adapts past successful solutions for code generation.

What are agents?

LangChain: *"… a system that uses an LLM to decide the control flow of an application." (Harrison Chase from LangChain wrote a great article on "What does it mean to be agentic?")*

NVIDIA: *"… a system that can use an LLM to reason through a problem, create a plan to solve the problem, and execute the plan with the help of a set of tools."*

AWS: *"… a software program that can interact with its environment, collect data, and use the data to perform self-determined tasks to meet predetermined goals."*

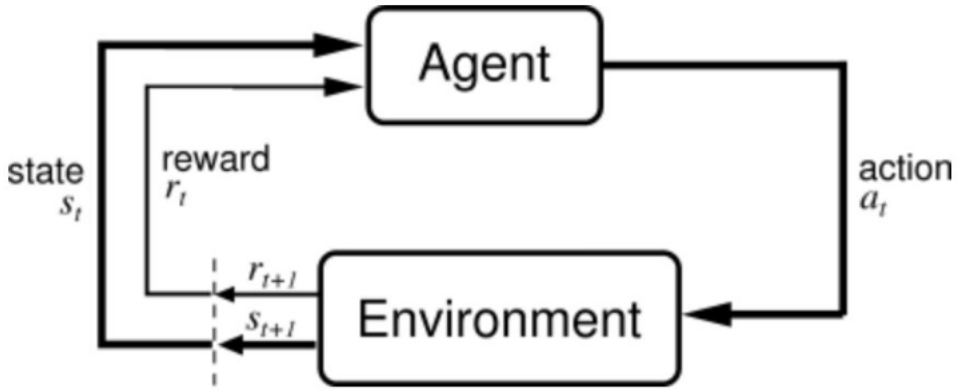


ref: https://medium.com/@alfredosone/ai-agents-how-to-build-digital-workers-4fe68bb20348

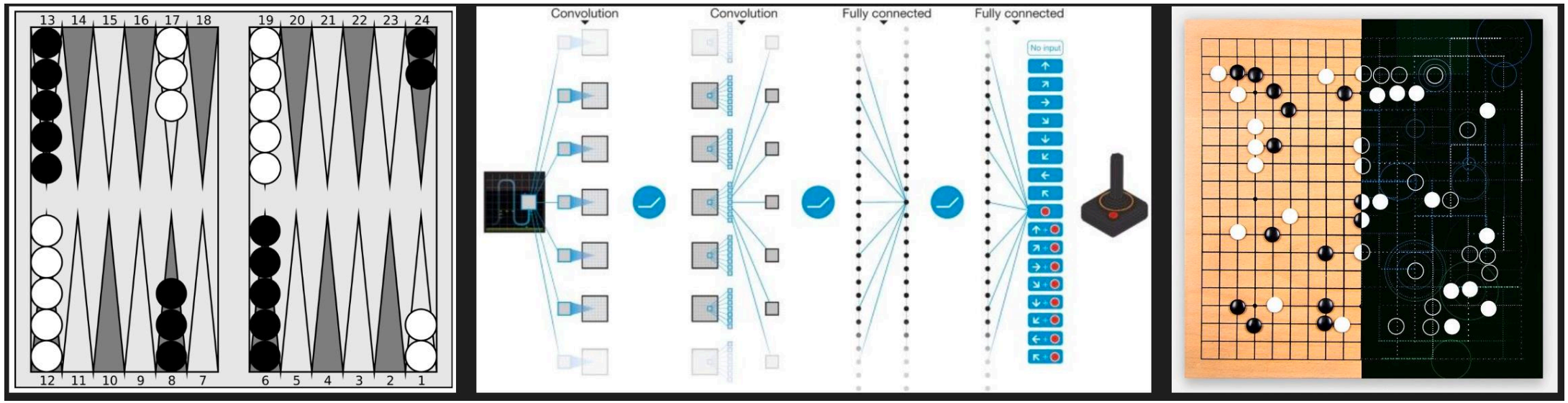
Agent in Reinforcement Learning

- The learner and decision-maker is called the Agent
- Agent interacts with an environment through actions
- Environment presents new states and rewards

A learning agent must be able to sense the state of its environment to some extent and must be able to take actions that affect the state. The agent also must have a goal or goals relating to the state of the environment.
—Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction, p. 2



Agent in Reinforcement Learning

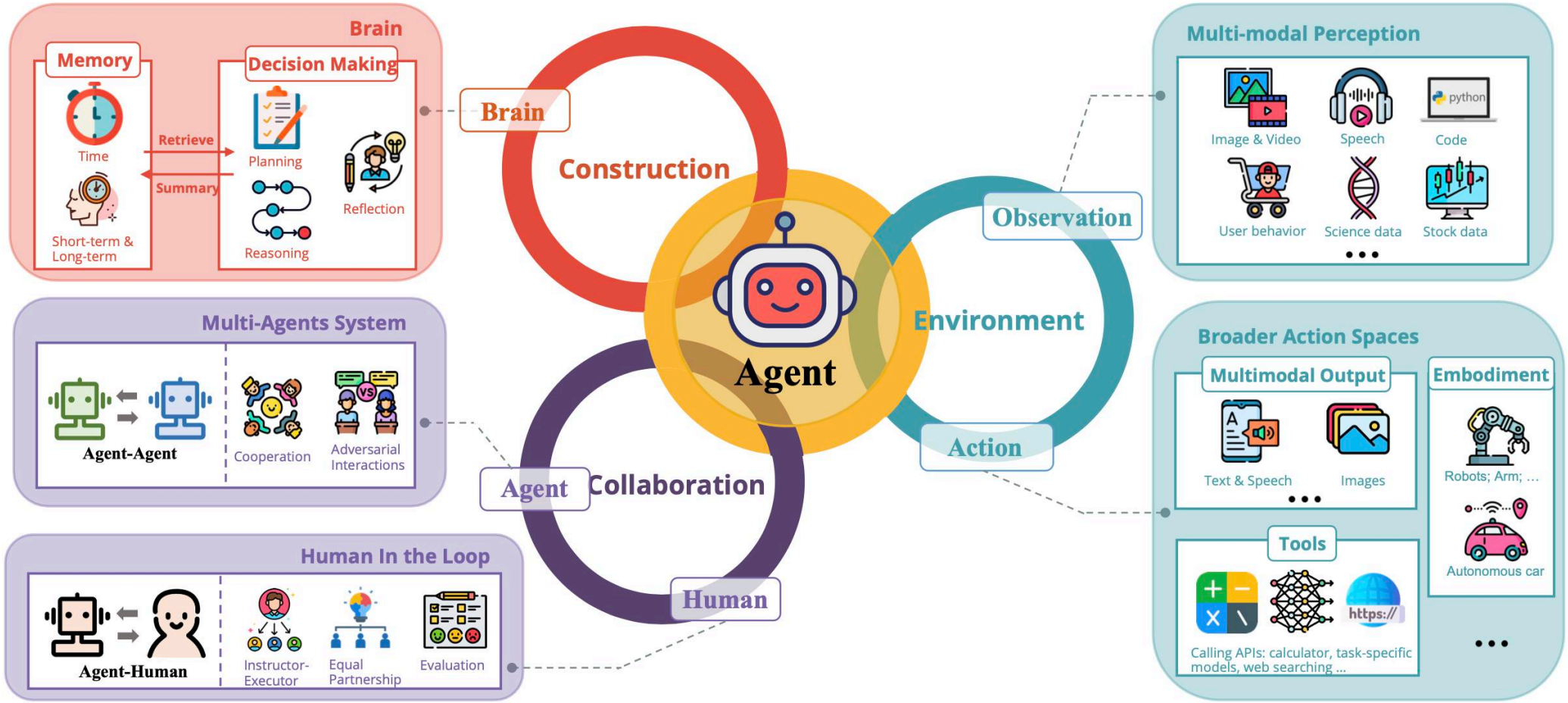


Backgammon (Q learning)

Atari Game (Deep Q Network)

Go Game (AlphaGo)

What are agents?



What are agents?

- Take actions according to observations

Action

- call external **APIs** for extra information that is missing from the model weights (often hard to change after pre-training): **Generating multimodal outputs; Embodied Action; Learning tools; Using tools; Making tools; ……**

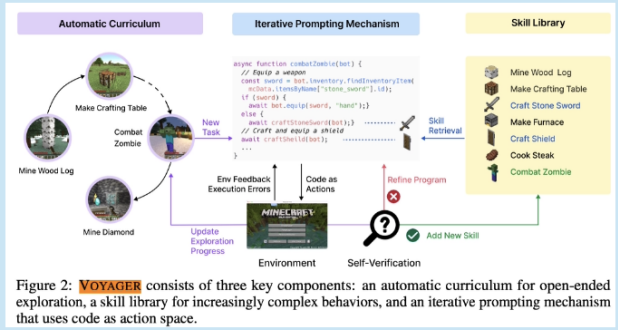
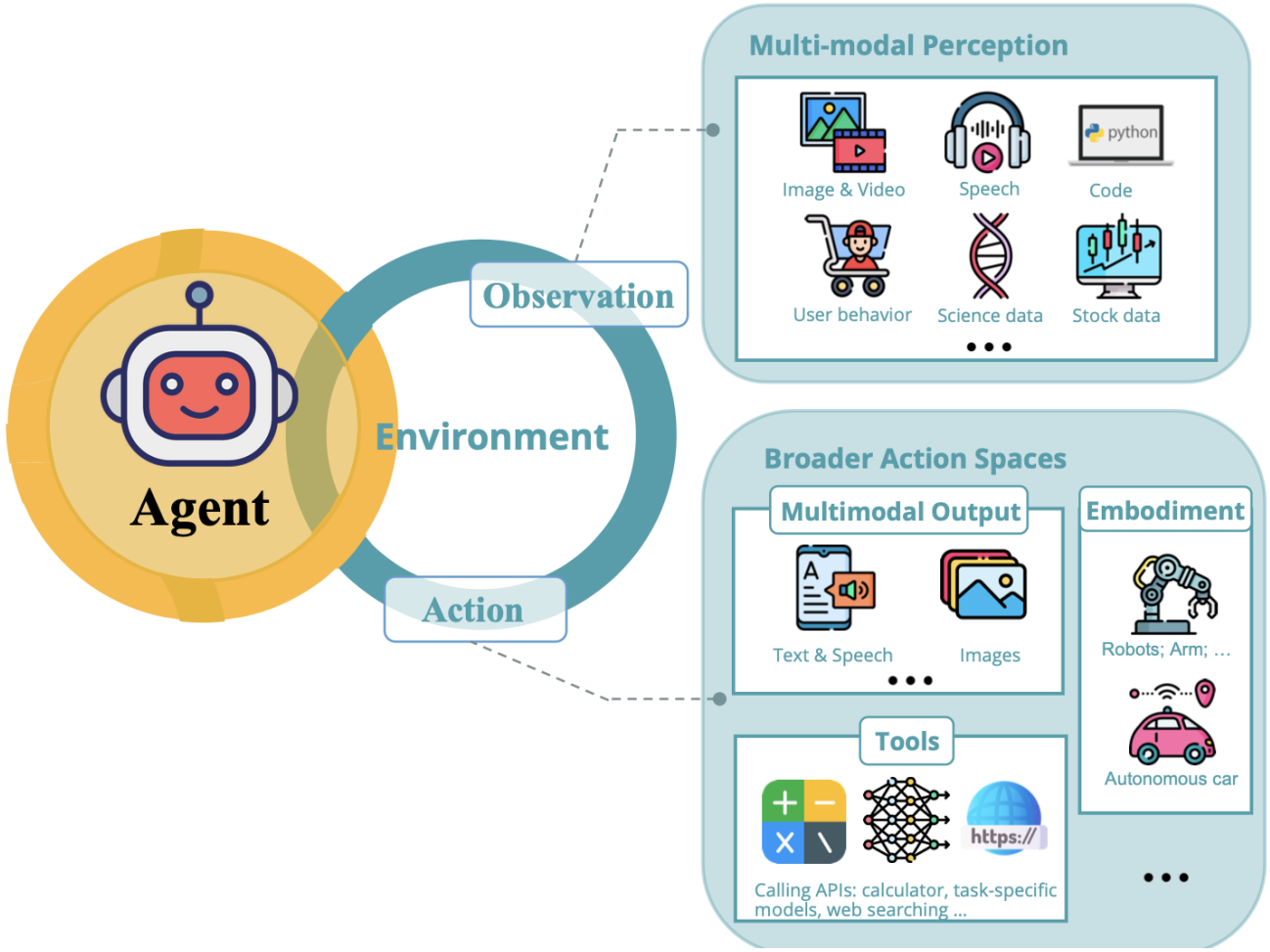
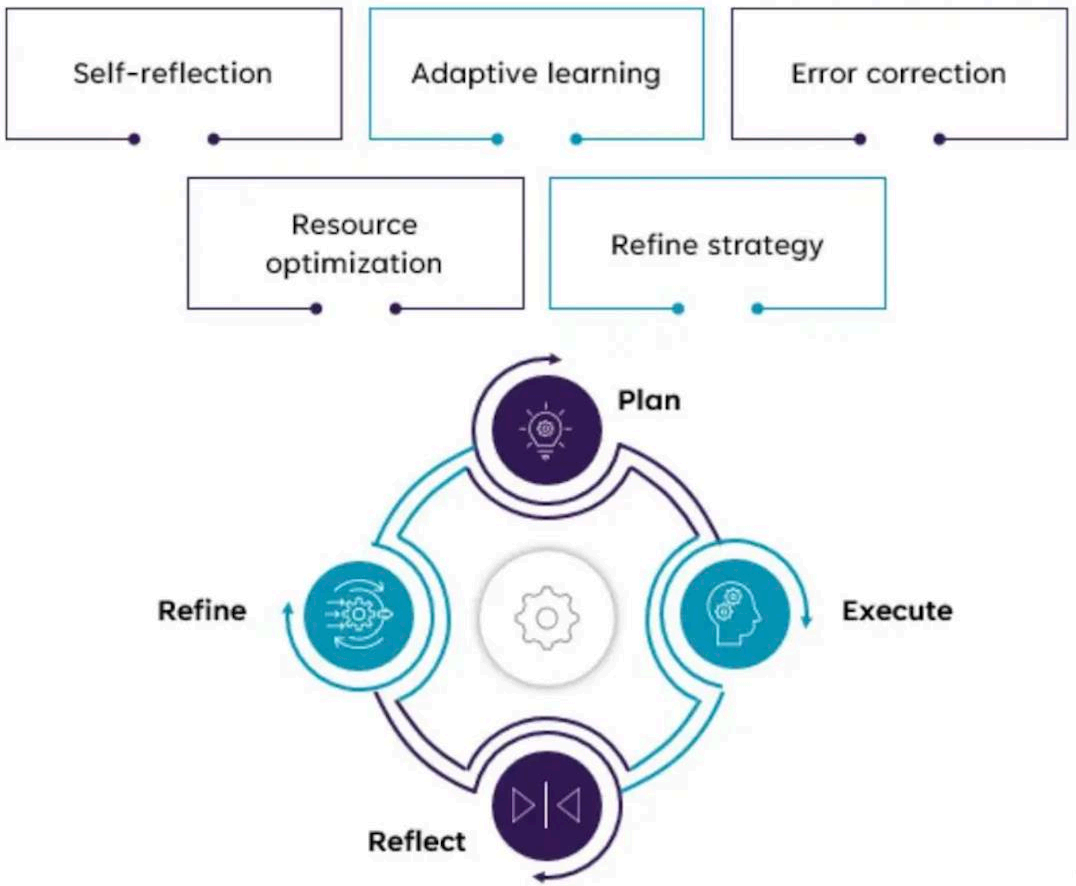
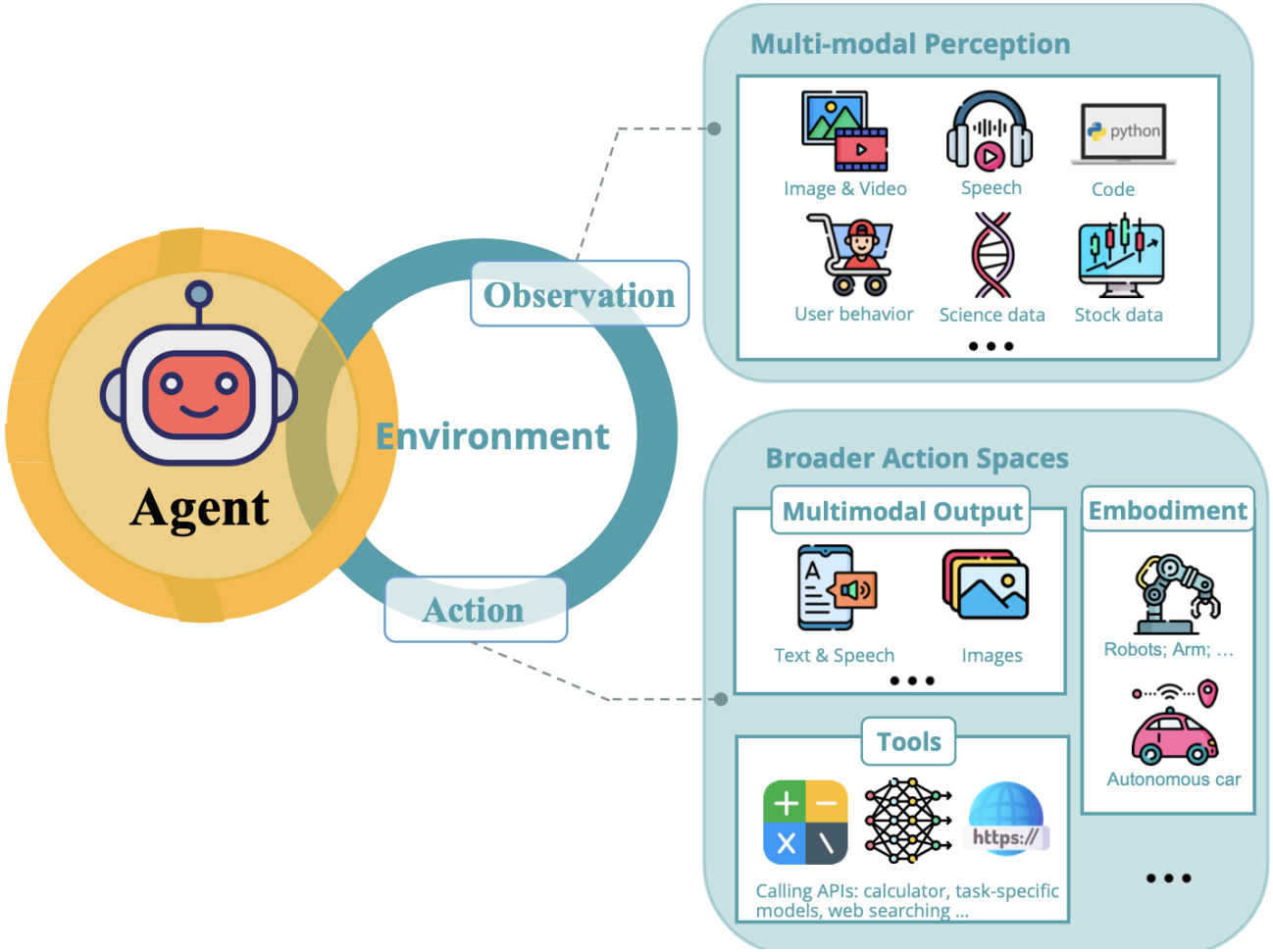


Figure 2: VOYAGER consists of three key components: an automatic curriculum for open-ended exploration, a skill library for increasingly complex behaviors, and an iterative prompting mechanism that uses code as action space.

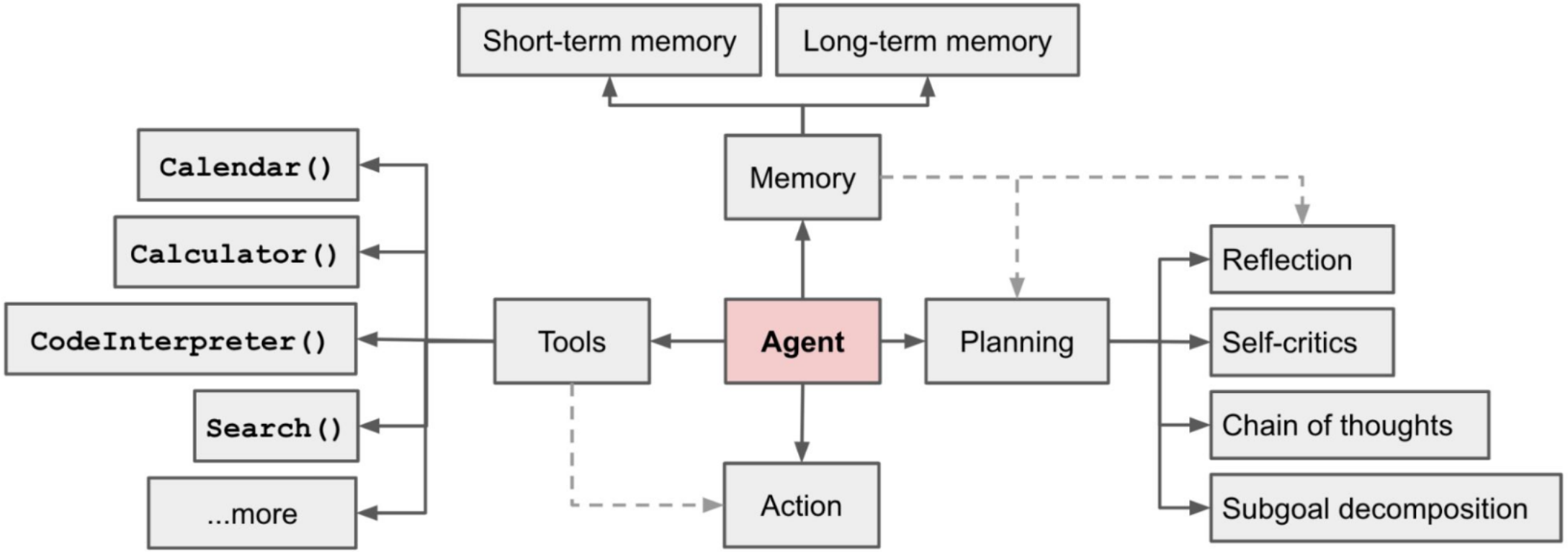


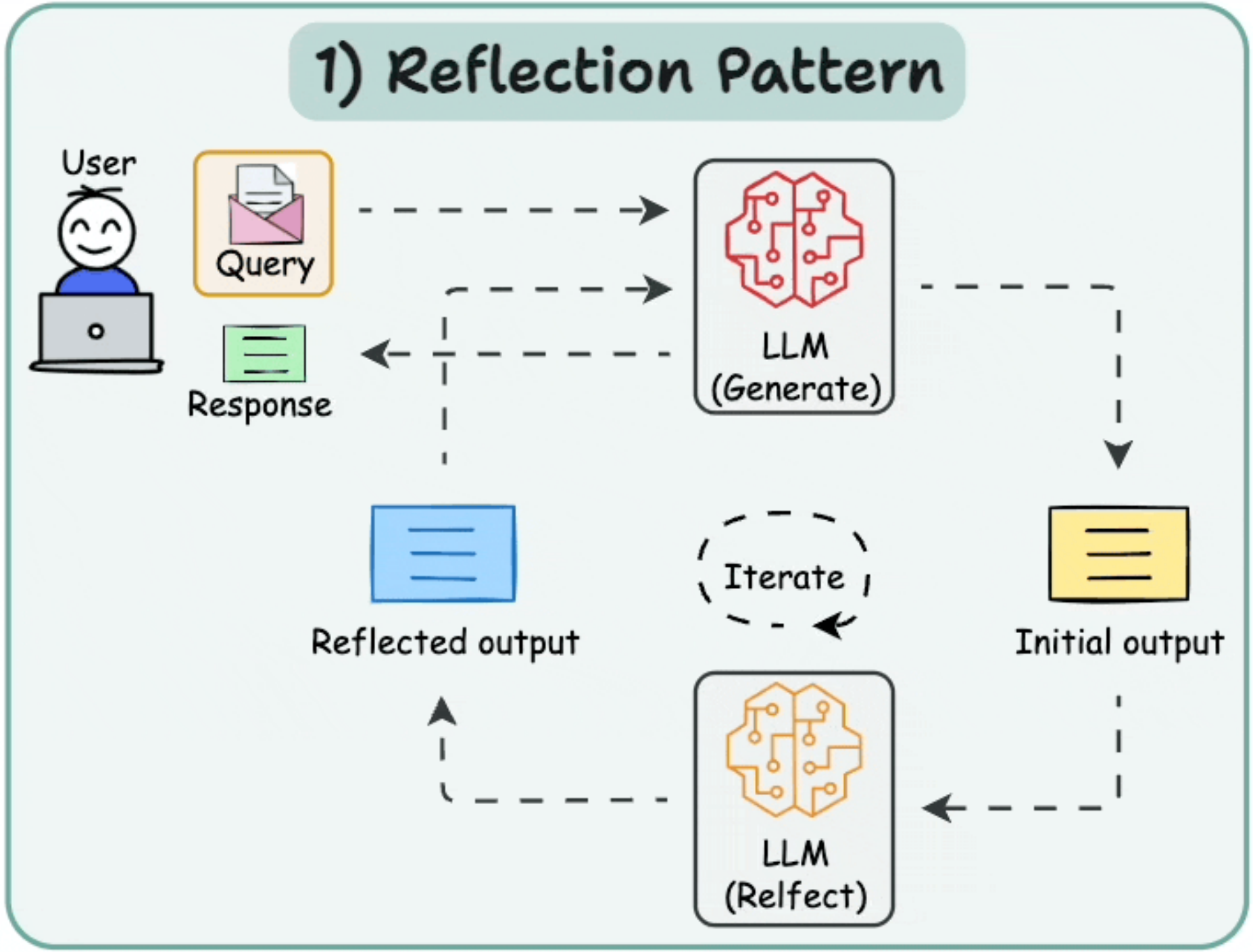
What are agents? Reflective



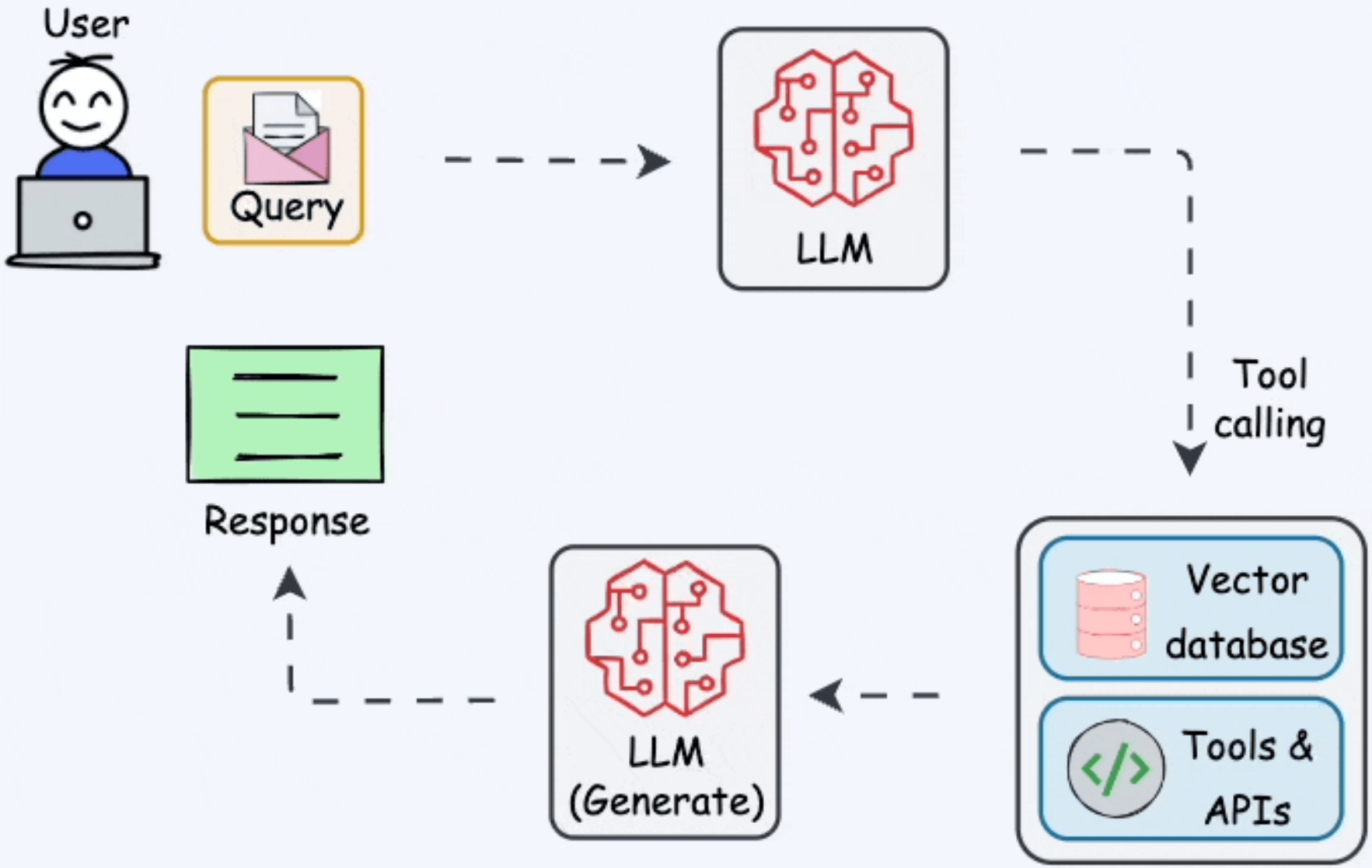


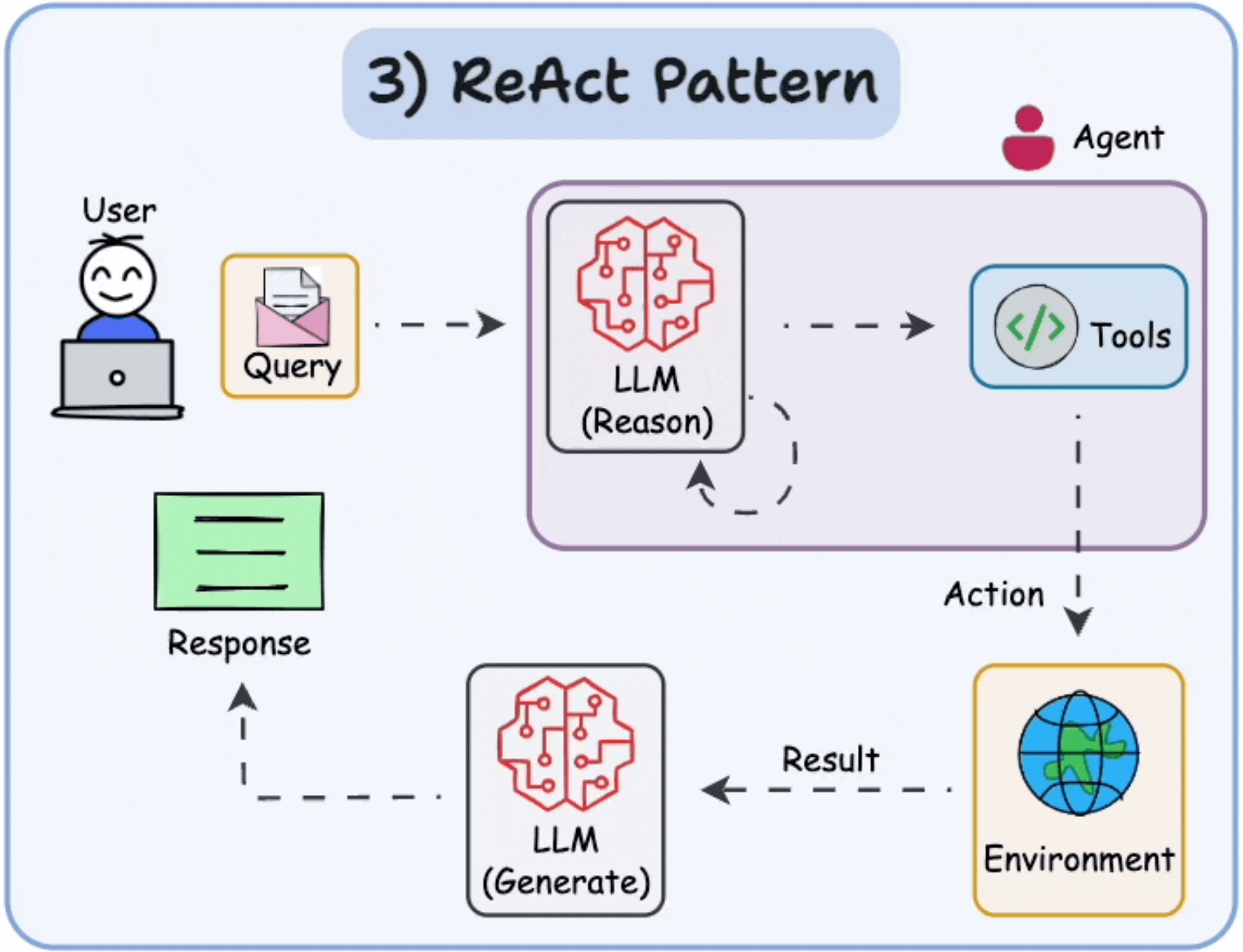
How to let LLM become an agent




1) Reflection Pattern
User
Query
Response
LLM
(Generate)
Reflected output
Iterate
Initial output
LLM
(Relfect)

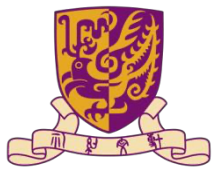
2) Tool Use Pattern




3) ReAct Pattern
Agent
User
Query
LLM
(Reason)
Tools
Action
Response
LLM
(Generate)
Result
Environment

Example ReAct prompt for Tool Selections

- You are an agent that answers questions by using two actions:
  - - search[query]: Google search the query. You can also use it to calculate math.
  - - finish[answer]: return the answer
- Your generation should have the following format:
  - Thought: to analyze the observation and inform the action.
  - Action: your action
- Question: If I have seven trillion dollars today, can I buy Apple, Nvidia, and Microsoft? If not, how much more money do I need?



Reasoning Problems in LLM

- Can we understand how the LLM approach the questions?

Q: If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?

A: The answer is **5**

Arithmetic Reasoning (AR)
(+ − ×÷…)

Q: Take the last letters of the words in "Elon Musk" and concatenate them

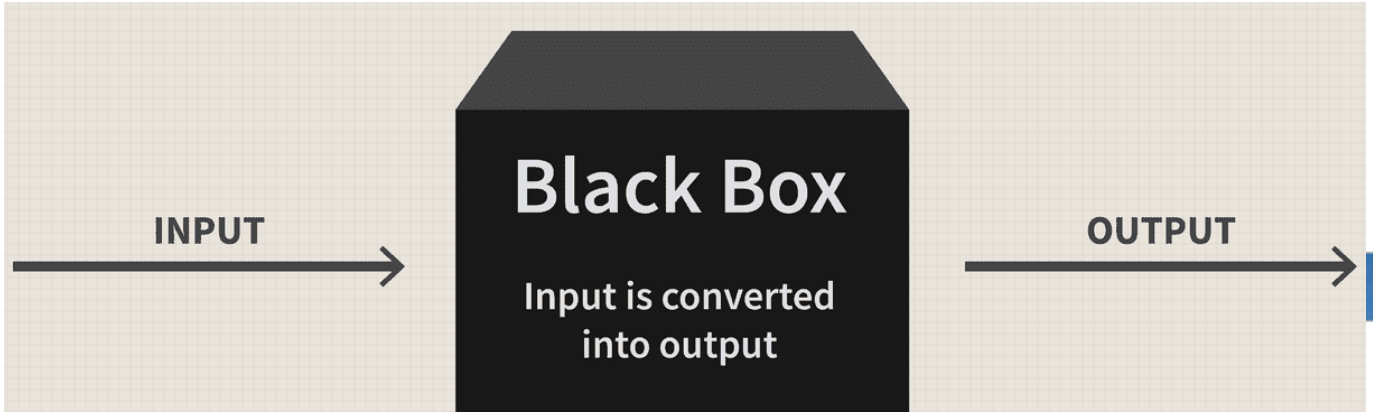
A: The answer is **nk.**

Symbolic Reasoning (SR)

Q: What home entertainment equipment requires cable? Answer Choices: (a) radio shack (b) substation (c) television (d) cabinet

A: The answer is **(c).**

Commonsense Reasoning (CR)

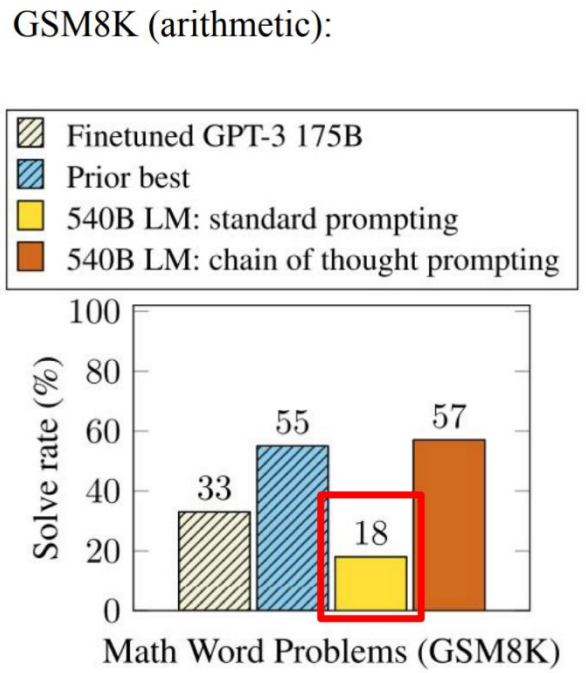


Reasoning Problems in LLM

- Few-shot standard prompting with even larger model (PaLM 540B) also does not work well.

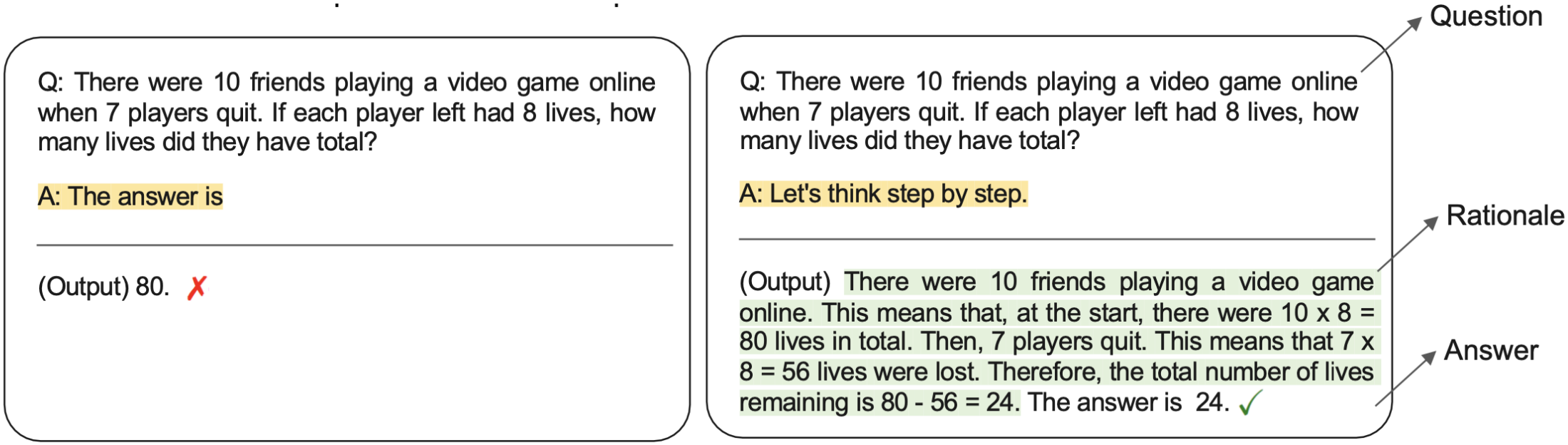
- Scaling up language model size does not efficiently achieve high performances, for Arithmetic Reasoning (AR), CommonSense Reasoning (CR) and Symbolic Reasoning (SR) tasks.

- Proposed solution: chain of thought prompting



What is Chain-of-Thought (CoT)?

- A chain of thought is a series of intermediate natural language reasoning steps that lead to the final output



Standard Format: <input → output>

CoT Format: <input → rationale → output>



What is Chain-of-Thought (CoT)? Benefits

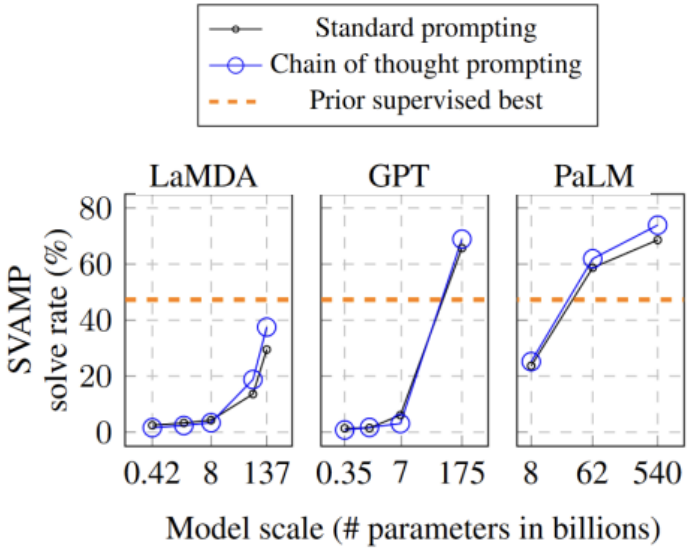
- Decomposition -> easier intermediate problems
- Interpretable
- More general than neural symbolic computing
- Leveraging prompting of LLM
- No update requires



LLMs are Strong CoT Reasoners

- LLMs show emergent abilities of solving challenging reasoning problems

Model	MultiArith		GSM8K		AddSub		AQUA-RAT		SingleEq		SVAMP	
	N/A	CoT	N/A	CoT	N/A	CoT	N/A	CoT	N/A	CoT	N/A	CoT
Zero-Shot Performance												
text-davinci-002	22.7	78.7	12.5	40.7	77.0	74.7	22.4	33.5	78.7	78.7	58.8	63.7
text-davinci-003	24.2	83.7	12.6	59.5	87.3	81.3	**28.0**	40.6	82.3	86.4	64.7	73.6
ChatGPT	**30.3**	**96.0**	**14.7**	**75.4**	**89.6**	**89.9**	23.6	**47.6**	**83.1**	**91.3**	**68.1**	**82.8**
Few-Shot Performance												
UL2	5.0	10.7	4.1	4.4	18.5	18.2	20.5	23.6	18.0	20.2	10.1	12.5
LaMDA	7.6	44.9	6.5	14.3	43.0	51.9	25.5	20.6	48.8	58.7	29.5	37.5
text-davinci-002	33.8	91.7	15.6	46.9	83.3	81.3	24.8	35.8	82.7	86.6	65.7	68.9
Codex	**44.0**	**96.2**	**19.7**	**63.1**	90.9	90.9	**29.5**	**45.3**	**86.8**	**93.1**	**69.9**	76.4
PaLM	42.2	94.7	17.9	56.9	**93.9**	**91.9**	25.2	35.8	86.5	92.3	69.4	**79.0**



[1] Qin, C., Zhang, A., Zhang, Z., Chen, J., Yasunaga, M. and Yang, D., 2023. Is ChatGPT a General-Purpose Natural Language Processing Task Solver?. arXiv preprint arXiv:2302.06476.

[2] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E.H., Le, Q.V. and Zhou, D., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. In *Advances in Neural Information Processing Systems* (NeurIPS). 2022.

Few-Shot-CoT

- In-context learning method by demonstrating **step-by-step reasoning exemplars (demonstrations)**

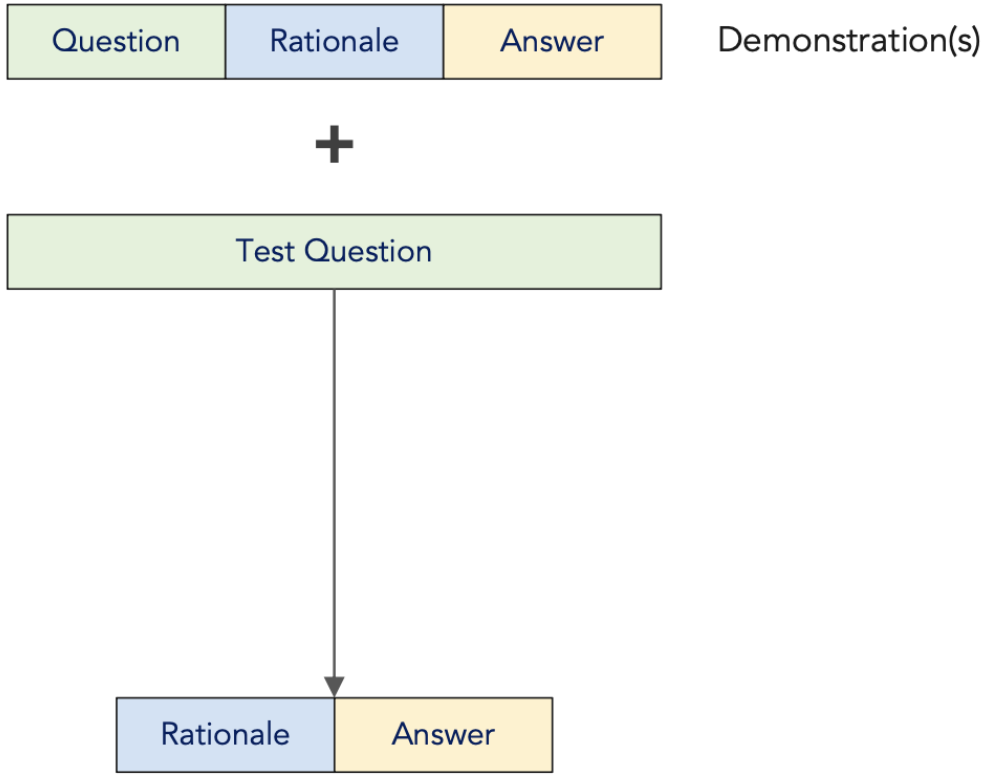
Q: There are 15 trees in the grove. Grove workers will plant trees in the grove today. After they are done, there will be 21 trees. How many trees did the grove workers plant today?
A: There are 15 trees originally. Then there were 21 trees after some more were planted. So there must have been 21 - 15 = 6. The answer is 6.
Q: A pet store had 64 puppies. In one day they sold 28 of them and put the rest into cages with 4 in each cage. How many cages did they use?
A:

↓

LLM

↓

A: The pet store had 64 puppies. They sold 28 of them. So they had 64 - 28 = 36 puppies left. They put them into cages with 4 in each cage. So they used 36 / 4 = 9 cages. The answer is 9.



Ref: Kojima, T., Gu, S.S., Reid, M., Matsuo, Y. and Iwasawa, Y., Large Language Models are Zero-Shot Reasoners. In Advances in Neural Information Processing Systems (NeurIPS). 2022.

Example ReAct prompt for Tool Selections

LLM

(GPT-4)

Thought: I need to find the current market capitalization of Apple, Nvidia, and Microsoft to understand how much it would cost to buy all of them.
Action: search["current market capitalization of Apple, Nvidia, and Microsoft"]

Observation: "Nvidia's market cap stood at $1.832 trillion as of about 10 a.m. ET Wednesday, while Alphabet's was at $1.814 trillion, bringing Nvidia behind just Microsoft (MSFT) at $3.015 trillion and Apple (AAPL) at $2.836 trillion."



(Mar 2024)

Thought: Now that I have the market cap for all companies, let's add them together and see if $7 trillion is enough.
Action: search["1.832 trillion + 3.015 trillion + 2.836 trillion"]

Observation: 7683000000000

Thought: The amount of money needed in addition to the $7 trillion is $683 billion.
Action: finish["You need an additional $683 billion to be able to buy Apple, Nvidia, and Microsoft."]

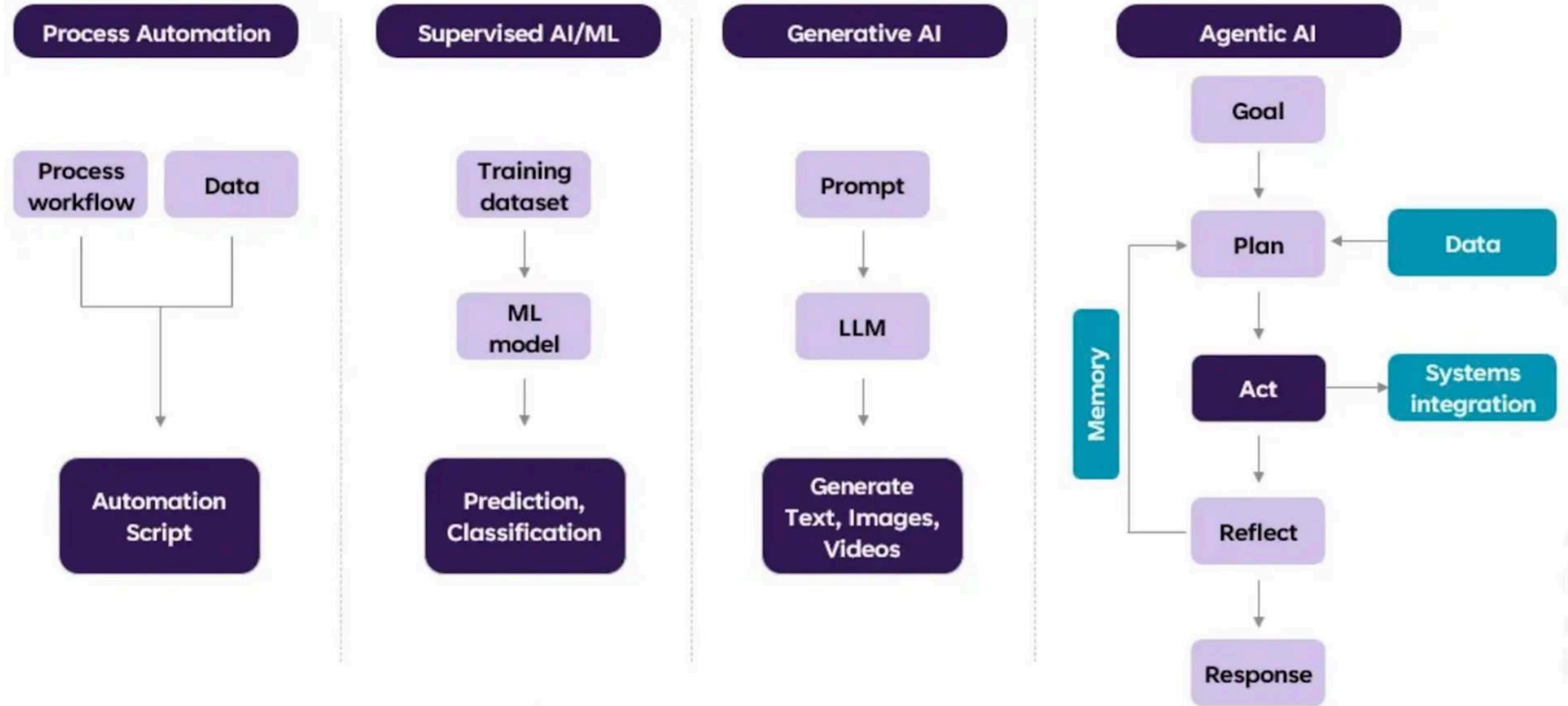




Example ReAct prompt for Tool Selections

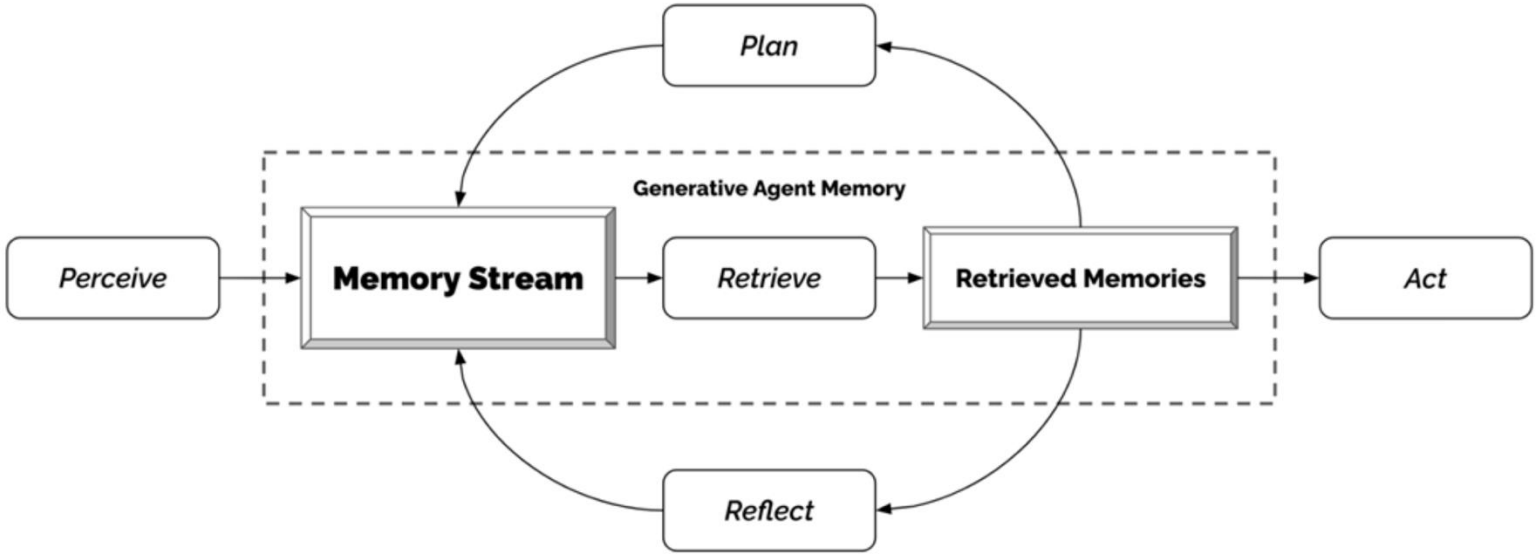


Synergy: acting support reasoning, reasoning guides acting


Process Automation
Process workflow
Data
Automation Script
Supervised AI/ML
Training dataset
ML model
Prediction, Classification
Generative AI
Prompt
LLM
Generate Text, Images, Videos
Agentic AI
Goal
Plan
Data
Memory
Act
Systems integration
Reflect
Response

Generative agents: the need for memory

- Context window cannot possibly hold all the event streams
- Even if possible, might be hard to attend to relevant events, or digest over them





Types of Memory

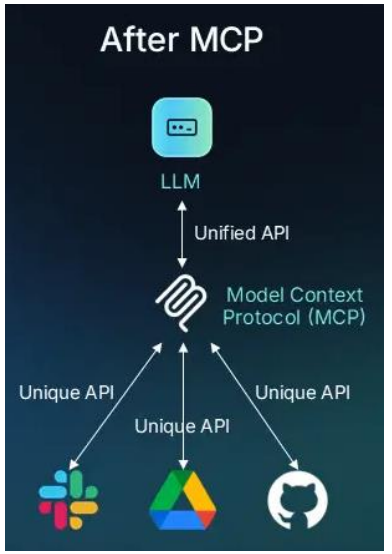
Types of Memory	
Episodic	Write: append-only event streams Read: retrieval based on heuristic scores
Semantic	Write: LLM reasoning over events Read: retrieval
Procedural	Write: Code-based skills Read: Embedding retrieval



Tools: MCP(model context protocol)

- an open source protocol that enables the seamless integration AI apps and agents with tools and data sources






```python
from mcp.server import MCPServer

class ExampleMCPServer(MCPServer):
    def list_resources(self, params):
        return {"documents": ["file1.txt", "file2.pdf"]}

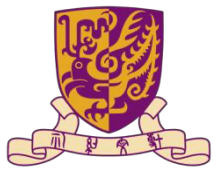
    def run_tool(self, tool_name, params):
        if tool_name == "get_weather":
            return {"weather": "Sunny, 72F"}
        return {"error": "Tool not found"}
server = ExampleMCPServer()
server.start()
```

```python
response = ai_model.generate(
    prompt="Summarize this document:",
    context=mcp_client.request("list_resources", {})
)
print(response)
```

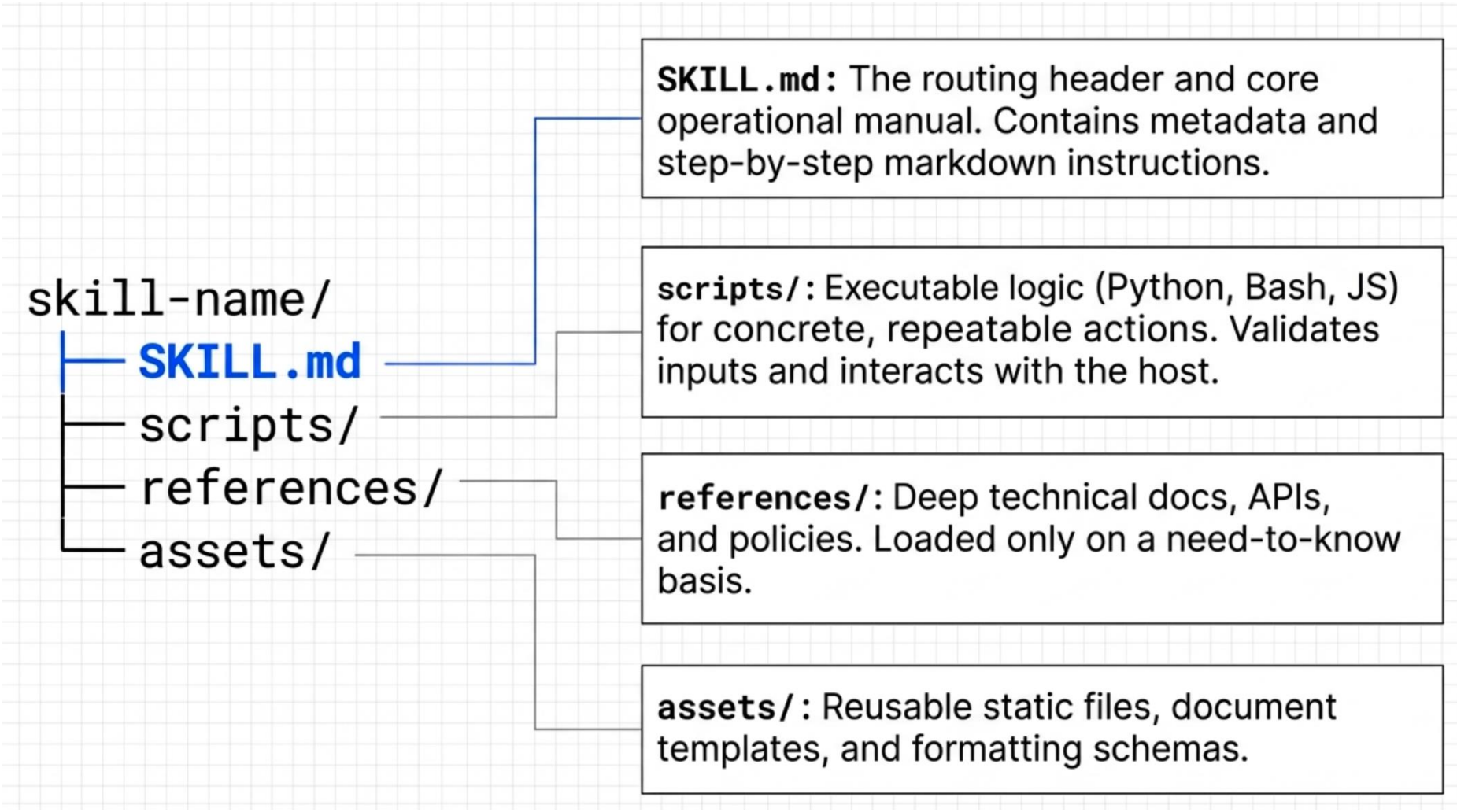
https://medium.com/data-science-in-your-pocket/best-mcp-servers-you-should-know-019226d01bca

Agent Skills: Concept

Concept	Definition
Tool	Atomic callable function. Primitive, one-off action.
Prompt	One-off instructions. Transient, bound to a single session.
Memory	Stored facts or episodic traces. assive, not an executable procedure.
Policy	Implicit decision rule. Black box, temporally extended but lacks explicit modularity.
Skill	Reusable procedural capability. Encapsulates when to trigger, how to execute, when to stop, and how to call.

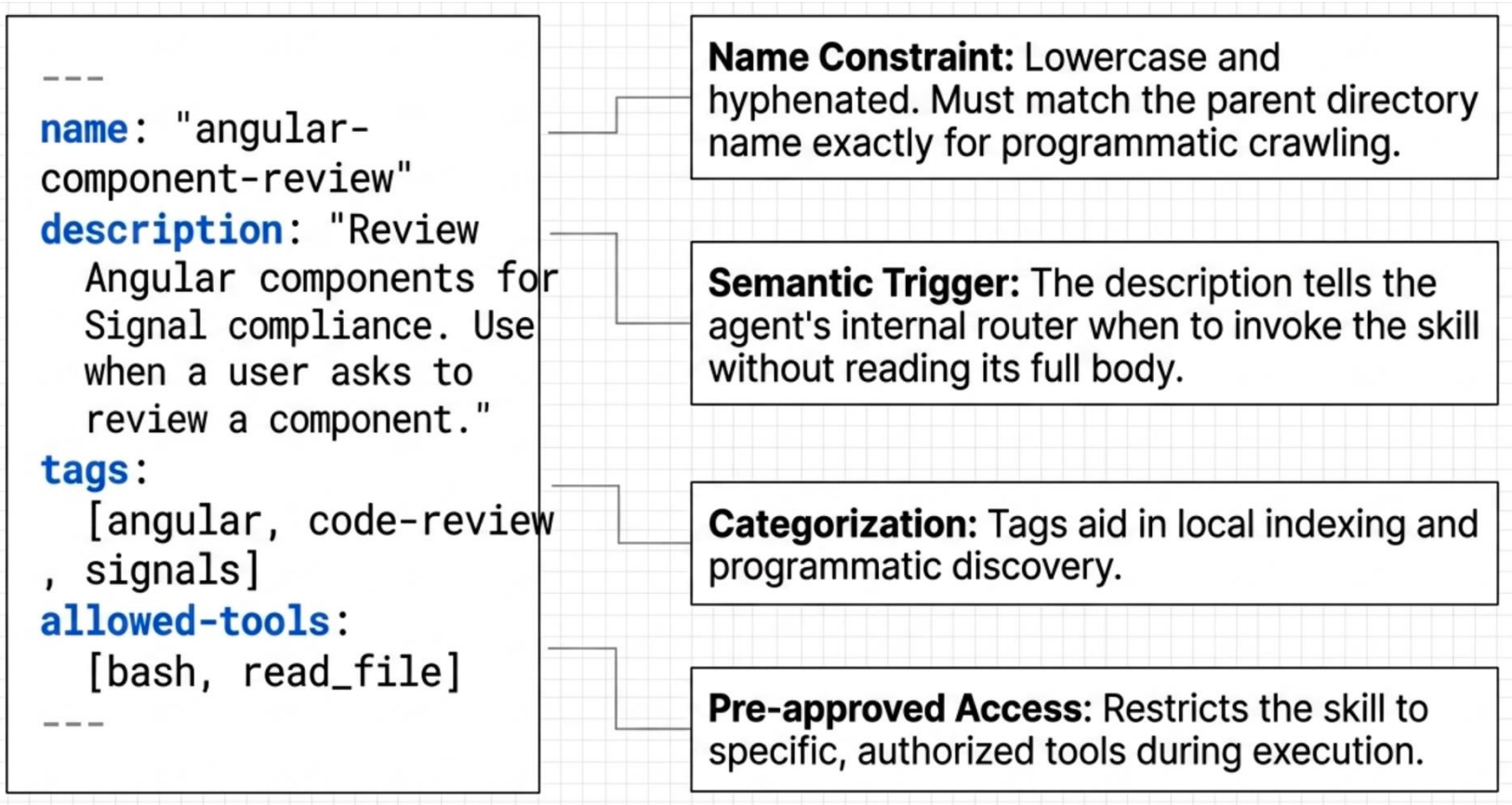


What a skills look likes





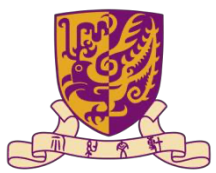
What a skills look likes





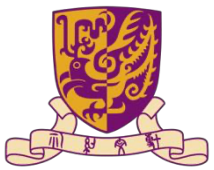
WHEN TO USE AGENTS, WHEN TO SKIP THEM

Situation	What to do
Short task, 2–4 steps, low stakes	Agents are fine. Monitor closely.
Irreversible actions (delete, pay, send)	Human confirmation before every step. Non-negotiable.
Multi-agent pipeline, 5+ handoffs	Validate every handoff. Budget for a 16% on-time rate.
Need consistent output every time	Use deterministic code. Agents aren't the right tool.
Research, synthesis, non-critical output	Agents are strong here. Iteration is tolerated.
Production database or file system access	Sandbox it. No direct access without rollback.
"The demo works, let's ship"	Don't. Run on real inputs for two weeks first.



Demo

- Using prompts to improve the generations



Questions?

Email: pakmingcheung@cuhk.edu.hk

